1. Fundamental konstantalar va olchovli tahlil (Fundamental Constants and Dimensional Analysis)

Fizikada har qanday tenglik munosabatida tenglamaning ikkala tomoni bir xil turdagi, yani bir xil olchovlikka (dimensions) ega bolishi kerak. Masalan, tenglamaning ong tomonidagi kattalik uzunlikni (length) ifodalab, chap tomonidagi kattalik vaqt oraligini (time interval) ifodalaydigan holat bolishi mumkin emas. Bu faktdan foydalanib, bazan fizik munosabatning shaklini masalani analitik yechmasdan deyarli chiqarish mumkin. Masalan, h balandlikdan erkin tushayotgan jismning g doimiy tortishish tezlanishi (gravitational acceleration) ostida tushish vaqtini topish soralganda, g va h kattaliklaridan vaqt oraligini ifodalovchi kattalik qurish kerak va buning mumkin bolgan yagona usuli $T = a(h/g)^{1/2}$ dir. Etibor bering, bu yechim hali aniqlanmagan olchamsiz (dimensionless) a koeffitsiyentini oz ichiga oladi va uni bu usul yordamida aniqlab bolmaydi. Bu koeffitsiyent 1, $\sqrt{3}$, π yoki boshqa haqiqiy son bolishi mumkin. Fizik munosabatlarni chiqarishning bu usuli olchovli tahlil (dimensional analysis) deb ataladi. Olchovli tahlilda olchamsiz koeffitsiyentlar muhim emas va ularni yozish shart emas. Yaxshiyamki, kopgina fizik masalalarda bu koeffitsiyentlar 1 atrofida boladi va ularni chiqarib tashlash fizik kattaliklarning kattalik tartibini (order of magnitude) ozgartirmaydi. Shuning uchun yuqoridagi masalaga olchovli tahlilni qollab, $T = (h/g)^{1/2}$ ni olamiz.

Umuman, fizik kattalikning olchovligi (dimensions) tortta fundamental kattalik: M (massa – mass), L (uzunlik – length), T (vaqt – time) va K (harorat – temperature) olchovlari orqali yoziladi. Ixtiyoriy x kattalikning olchovligi $[x]$ bilan belgilanadi. Misol tariqasida v tezlik (velocity), E_k kinetik energiya (kinetic energy) va C_v issiqlik sigimi (heat capacity) ning olchovliklarini ifodalaymiz: $[v] = LT^{-1}$, $[E_k] = ML^2T^{-2}$, $[C_v] = ML^2T^{-2}K^{-1}$.

1.1 Fundamental konstantalarning (fundamental constants) – Plank doimiysi h (Planck's constant), yoruglik tezligi c (speed of light), universal tortishish doimiysi G (universal constant of gravitation) va Boltsman doimiysi k_B (Boltzmann constant) – olchovliklarini uzunlik, massa, vaqt va harorat olchovlari orqali toping. (0.8 ball)

1.2 Stefan–Boltsman qonuni (Stefan-Boltzmann law) qora jismning emissiv quvvati (emissive power) – qora jismning birlik sirtidan birlik vaqtda chiqargan umumiy energiyasi – $\sigma\theta^4$ ga teng ekanligini aytadi, bu yerda σ – Stefan–Boltsman doimiysi (Stefan-Boltzmann's constant), θ – qora jismning absolyut harorati (absolute temperature). Stefan–Boltsman doimiysining olchovligini uzunlik, massa, vaqt va harorat olchovlari orqali aniqlang. (0.5 ball)

1.3 Stefan–Boltsman doimiysi fundamental konstanta emas va uni fundamental konstantalar orqali yozish mumkin, yani $\sigma = ah^\alpha c^\beta G^\gamma k_B^\delta$ yozish mumkin. Bu munosabatda a – 1 atrofidagi olchamsiz parametr. Yuqorida aytib otilganidek, a ning aniq qiymati bizning nuqtai nazarimizdan muhim emas, shuning uchun uni 1 ga tenglaymiz. Olchovli tahlildan foydalanib α, β, γ va δ ni toping. (1.0 ball)

2. Qora tuynuklar fizikasi (Physics of Black Holes)

Ushbu qismda biz olchovli tahlil yordamida qora tuynuklarning (black holes) bazi xususiyatlarini aniqlamoqchimiz. Fizikadagi "sochsizlik teoremasi" (no hair theorem) deb nomlanuvchi teoremaga kora, biz ushbu masalada korib chiqayotgan qora tuynukning barcha xarakteristikalari faqat qora tuynuk massasiga bogliq. Qora tuynukning xarakteristikalaridan biri bu uning hodisa ufqi (event horizon) maydoni (area). Taxminan aytganda, hodisa ufqi qora tuynukning chegarasidir. Bu chegara ichida tortishish shunchalik kuchliki, hatto yoruglik ham chegara bilan chegaralangan hududdan chiqa olmaydi.

Biz qora tuynuk massasi m bilan uning hodisa ufqi maydoni A ortasidagi munosabatni topmoqchimiz. Bu maydon qora tuynuk massasiga, yoruglik tezligiga va universal tortishish doimiysiga bogliq. 1.3 qismdagi kabi $A = G^\alpha c^\beta m^\gamma$ yozamiz.

2.1 Olchovli tahlildan foydalanib α, β va γ ni toping. (0.8 ball)

2.2 Entropiyaning (entropy) termodinamik tarifi $dS = dQ/\theta$ dan foydalanib (dQ – almashinuvchi issiqlik, θ – tizimning absolyut harorati), entropiyaning olchovligini toping. (0.2 ball)

2.3 1.3 qismdagidek, olchamli η konstantasini fundamental konstantalar h, c, G va k_B ning funksiyasi sifatida ifodalang. (1.1 ball)

Masalaning qolgan qismlari uchun olchovli tahlildan foydalanmang, lekin oldingi bolimlarda olingan natijalardan foydalanishingiz mumkin.

3. Xoking nurlanishi (Hawking Radiation)

Yarim kvant mexanik yondashuv bilan Xoking (Hawking) klassik nuqtai nazarga zid ravishda, qora tuynuklar qora jism nurlanishiga oxshash nurlanish chiqarishi va bu nurlanish Xoking harorati (Hawking temperature) deb ataladigan haroratga ega ekanligini takidladi.

3.1 $E = mc^2$ (qora tuynuk energiyasini uning massasi orqali beradi) va termodinamika qonunlaridan foydalanib, qora tuynukning Xoking harorati θ_H ni uning massasi va fundamental konstantalar orqali ifodalang. Qora tuynuk atrofga ish bajarmaydi deb faraz qiling. (0.8 ball)

3.2 Shunday qilib, ajratilgan qora tuynukning massasi Xoking nurlanishi tufayli ozgaradi. Qora tuynukning massasining ozgarish tezligini (rate of change) uning Xoking harorati θ_H ga bogliqligini topish uchun Stefan–Boltsman qonunidan foydalaning va natijani qora tuynuk massasi va fundamental konstantalar orqali ifodalang. (0.7 ball)

3.3 Massasi m bolgan ajratilgan qora tuynuk toliq buglanishi (yani butun massasini yoqotishi) uchun ketadigan vaqt t^* ni toping. (0.8 ball)

3.4 Qora tuynukning issiqlik sigimini (heat capacity) toping. (0.4 ball)

4. Qora tuynuklar va kosmik fon nurlanishi (Black Holes and the Cosmic Background Radiation)

Kosmik fon nurlanishiga (cosmic background radiation) tasirlanayotgan qora tuynukni korib chiqaylik. Kosmik fon nurlanishi butun olamni toldiruvchi θ_B haroratli qora jism nurlanishidir. Umumiy maydoni A bolgan jism, shuning uchun, vaqt birligida $\sigma\theta_B^4 \times A$ ga teng energiyani oladi. Demak, qora tuynuk Xoking nurlanishi orqali energiya yoqotadi va kosmik fon nurlanishidan energiya oladi.

4.1 Qora tuynuk massasining ozgarish tezligini (rate of change) qora tuynuk massasi, kosmik fon nurlanishining harorati va fundamental konstantalar orqali toping. (0.8 ball)

4.2 Malum bir m^* massada bu ozgarish tezligi nolga aylanadi. m^* ni toping va uni θ_B va fundamental konstantalar orqali ifodalang. (0.4 ball)

4.3 4.2 qismdagi javobingizdan foydalanib, 4.1 qismdagi javobingizda θ_B orniga qoying va qora tuynuk massasining ozgarish tezligini m, m^* va fundamental konstantalar orqali ifodalang. (0.2 ball)

4.4 Kosmik fon nurlanishi bilan termal muvozanatda (thermal equilibrium) bolgan qora tuynukning Xoking haroratini toping. (0.4 ball)

4.5 Bu muvozanat barqarormi (stable) yoki beqarormi (unstable)? Nima uchun? (Javobingizni matematik asoslang). (0.6 ball)

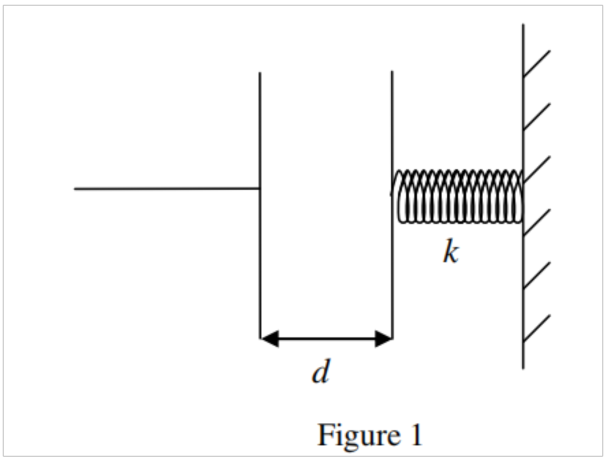
5. Elektromexanik akselerometr (Electromechanical Accelerometer)

Ushbu masalada biz avtomobillarning toqnashuvi vaqtida xavfsizlik havo yostiqlarini (safety air bags) ishga tushirish uchun moljallangan akselerometrlarning (accelerometers) soddalashtirilgan modeli bilan shugullanamiz. Tezlanish (acceleration) malum chegaradan oshganda, tizimning elektr parametrlaridan biri (masalan, zanjirning malum bir nuqtasidagi kuchlanish (voltage)) chegara qiymatidan oshib ketadigan va natijada havo yostigi ishga tushadigan elektromexanik tizim qurmoqchimiz.

Eslatma: Ushbu masalada tortishish kuchini (gravity) etiborsiz qoldiring.

1. Parallel plastinkali kondensator (Capacitor with parallel plates)

1-rasmda (Rasm (1).png) korsatilganidek, parallel plastinkali kondensatorni korib chiqaylik. Kondensatordagi har bir plastinkaning maydoni A va ikki plastinka orasidagi masofa d. Ikki plastinka orasidagi masofa plastinkalarning olchamlaridan ancha kichik. Ushbu plastinkalardan biri prujina (spring) orqali devorga tegib turadi (prujinaning prujina konstantasi k), ikkinchi plastinka esa qozgalmas (fixed). Plastinkalar orasidagi masofa d bolganda prujina na chozilgan, na siqilgan, boshqacha aytganda bu holatda prujinaga hech qanday kuch tasir qilmaydi. Havoning dielektrik singdiruvchanligi (permittivity) vakuumdagi ε_0 ga teng deb faraz qiling. Kondensator plastinkalarining bu oraliqdagi sigimi (capacitance) $C_0 = \varepsilon_0 A/d$ ga teng. Plastinkalarga $+Q$ va $-Q$ zaryadlar (charges) qoyamiz va tizimni mexanik muvozanatga (mechanical equilibrium) keltiramiz.



1-rasm (Rasm (1).png): Parallel plastinkali kondensator.

1.1 Plastinkalarning bir-biriga korsatadigan elektr kuchini (electrical force) F_E hisoblang. (0.8 ball)

1.2 Prujinaga ulangan plastinkaning siljishi (displacement) x bolsin. x ni toping. (0.6 ball)

1.3 Bu holatda kondensator plastinkalari orasidagi elektr potensiallar farqi (electrical potential difference) V ni Q, A, d, k orqali ifodalang. (0.4 ball)

1.4 Zaryadning potensiallar farqiga nisbati sifatida aniqlanadigan kondensator sigimi C bolsin. C/C_0 ni Q, A, d va k ning funksiyasi sifatida toping. (0.3 ball)

1.5 Tizimda saqlanayotgan umumiy energiyani (total energy) U ni Q, A, d va k orqali ifodalang. (0.6 ball)

2-rasm (Rasm (2).png) massasi M bolgan jismni korsatadi. U massasi cheksiz kichik deb hisoblanadigan otkazuvchi plastinkaga va ikkita bir xil prujina konstantasi k ga ega prujinaga biriktirilgan. Otkazuvchi plastinka ikkita qozgalmas otkazuvchi plastinka orasidagi boshliqda oldinga va orqaga harakatlana oladi. Barcha plastinkalar oxshash va bir xil A maydoniga ega. Shunday qilib, bu uchta plastinka ikkita kondensatorni tashkil qiladi. 2-rasmda korsatilganidek, qozgalmas plastinkalar V va $-V$ potensiallariga ulangan va orta plastinka ikki holatli kalit (two-state switch) orqali yerga (ground) ulangan. Harakatlanuvchi plastinkaga ulangan sim plastinkaning harakatiga xalaqit bermaydi va uchala plastinka doimo parallel bolib qoladi. Butun kompleks tezlanmayotganida (accelerated not being) har bir qozgalmas plastinkadan harakatlanuvchi plastinkagacha bolgan masofa d ga teng, bu plastinkalarning olchamlaridan ancha kichik. Harakatlanuvchi plastinkaning qalinligi etiborsiz qoldirilishi mumkin.

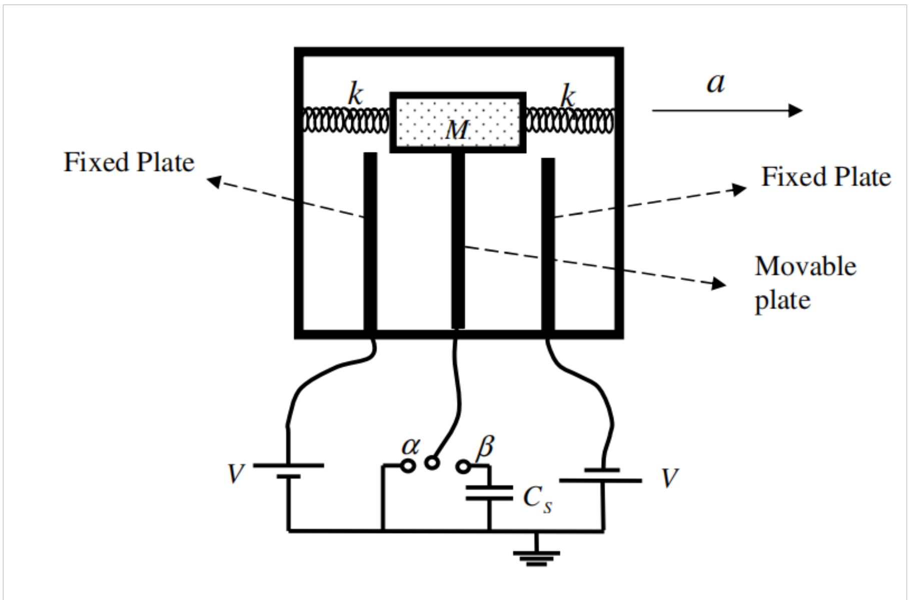


Figure 1: 2-rasm (Rasm (2).png): Harakatlanuvchi plastinkali akselerometr tizimi.

Kalit ikki holatdan birida bolishi mumkin: α va β. Kondensator kompleksi avtomobil bilan birga tezlanayapti deb faraz qiling va bu tezlanish (acceleration) ozgarmas (constant). Ushbu ozgarmas tezlanish davomida prujina tebranmaydi va bu murakkab kondensatorning barcha komponentlari oz muvozanat holatlarida, yani ular bir-biriga nisbatan harakat qilmaydi, shuning uchun avtomobilga nisbatan ham harakat qilmaydi.

Tezlanish tufayli harakatlanuvchi plastinka ikki qozgalmas plastinkaning ortasidan malum bir x miqdoriga siljiydi.

2. Kalit α holatida (Switch in state α)

Kalit α holatida, yani harakatlanuvchi plastinka sim orqali yerga ulangan holda:

2.1 Har bir kondensatordagi zaryadni (charge on each capacitor) x ning funksiyasi sifatida toping. (0.4 ball)

2.2 Harakatlanuvchi plastinkaga tasir qiluvchi aniq elektr kuchini (net electrical force) F_E ni x ning funksiyasi sifatida toping. (0.4 ball)

2.3 $d \gg x$ deb faraz qiling va x^2 tartibidagi hadlarni d^2 tartibidagi hadlarga nisbatan etiborsiz qoldirish mumkin. Oldingi qismdagi javobni soddalashtiring. (0.2 ball)

2.4 Harakatlanuvchi plastinkaga tasir qiluvchi umumiy kuchni (elektr va prujina kuchlarining yigindisi) $-k_{\text{eff}}x$ korinishida yozing va k_{eff} ning shaklini bering. (0.7 ball)

2.5 Ozgarmas tezlanish a ni x ning funksiyasi sifatida ifodalang. (0.4 ball)

3. Kalit β holatida (Switch in state β)

Endi kalit β holatida deb faraz qilaylik, ya'ni harakatlanuvchi plastinka sig'imi C_s bo'lgan kondensator (bunda kondensatorlarda dastlabki zaryad yo'q) orqali yerga ulangan. Agar harakatlanuvchi plastinka markaziy holatidan x miqdoriga siljigan bo'lsa,

3.1 C_s kondensatoridagi elektr potensiallar farqi (electrical potential difference) V_s ni x ning funksiyasi sifatida toping. (1.5 ball)

3.2 Yana $d \gg x$ deb faraz qiling va x^2 tartibidagi hadlarni d^2 tartibidagi hadlarga nisbatan e'tiborsiz qoldiring. Oldingi qismdagi javobingizni soddalashtiring. (0.2 ball)

4. Havo yostig'ini sozlash (Adjusting the air bag)

Biz masala parametrlarini shunday sozlamoqchimizki, havo yostig'i (air bag) normal tormozlashda ishga tushmasin, lekin to'qnashuv vaqtida haydovchining boshi old oyna yoki rulga urilishining oldini olish uchun yetarlicha tez ochilsin. 2-qismda ko'rganingizdek, prujinalar va elektr zaryadlari tomonidan harakatlanuvchi plastinkaga ta'sir qiluvchi kuchni effektiv prujina konstantasi k_{eff} bo'lgan prujinaning kuchi sifatida ifodalash mumkin. Butun kondensator kompleksi massasi M va prujina konstantasi k_{eff} bo'lgan massa-prujina tizimiga o'xshaydi, bu tizim o'zgarmas tezlanish a (bu masalada avtomobilning tezlanishi) ta'sirida.

Eslatma: Masalaning ushbu qismida massa va prujinaning o'zgarmas tezlanish ostida muvozanatda bo'lishi va shuning uchun avtomobilga nisbatan qo'zg'almasligi haqidagi faraz endi o'rinli emas.

Ishqalanishni e'tiborsiz qoldiring va masala parametrlarining quyidagi raqamli qiymatlarini hisobga oling:

$$d = 1.0\,\text{cm}, \quad A = 2.5\times10^{-2}\,\text{m}^2, \quad k = 4.2\times10^{3}\,\text{N/m}, \quad \varepsilon_0 = 8.85\times10^{-12}\,\text{C}^2/\text{Nm}^2, \quad V = 12\,\text{V}, \quad M = 0$$

4.1 Ushbu ma'lumotlardan foydalanib, 2.3 bo'limda hisoblagan elektr kuchining prujinalar kuchiga nisbatini toping va elektr kuchlarini prujina kuchlariga nisbatan e'tiborsiz qoldirish mumkinligini ko'rsating. (0.6 ball)

Kalit β holatida bo'lgan holda biz elektr kuchlarini hisoblamagan bo'lsak ham, bu holatda ham xuddi shunday elektr kuchlari kichik va ularni e'tiborsiz qoldirish mumkinligini ko'rsatish mumkin.

4.2 Agar avtomobil o'zgarmas tezlik bilan harakatlanayotganda to'satdan o'zgarmas a tezlanish bilan tormozlasa, harakatlanuvchi plastinkaning maksimal siljishi (maximum displacement) qancha bo'ladi? Javobingizni parametrlar orqali bering. (0.6 ball)

Kalit β holatida deb faraz qiling va tizim shunday yaratilganki, C_s kondensatoridagi kuchlanish $V_s = 0.15\,\mathrm{V}$ ga yetganda havo yostig'i ishga tushadi. Biz havo yostig'i avtomobilning tezlanishi tortishish tezlanishidan $g = 9.8\,\mathrm{m/s^2}$ kichik bo'lgan normal tormozlash vaqtida ishga tushmasligini, aks holda esa ishga tushishini xohlaymiz.

4.3 Ushbu maqsad uchun C_s qancha bo'lishi kerak? (0.6 ball)

Biz havo yostig'i haydovchining boshi old oyna yoki rulga urilishining oldini olish uchun yetarlicha tez ishga tushadimi yoki yo'qligini aniqlamoqchimiz. Faraz qiling, to'qnashuv natijasida avtomobil $8g$ ga teng sekinlanish (deceleration) ni boshdan kechiradi, ammo haydovchining boshi o'zgarmas tezlikda harakat qilishda davom etadi.

4.4 Haydovchining boshi va rul o'rtasidagi masofani taxmin qilib, haydovchining boshi rulga urilguncha o'tadigan vaqt t_1 ni toping. (0.8 ball)

4.5 Havo yostig'i ishga tushguncha o'tadigan vaqt t_2 ni toping va uni t_1 bilan solishtiring. Havo yostig'i vaqtida ishga tushadimi? Havo yostig'i bir zumda (instantaneously) ochiladi deb faraz qiling. (0.9 ball)

Ikki yulduz ularning massa markazi atrofida aylanib, qo'sh yulduz tizimini (binary star system) hosil qiladi. Galaktikamizdagi yulduzlarning deyarli yarmi qo'sh yulduz tizimlaridir. Bu yulduz tizimlarining ko'pchiligining ikkilamchi tabiatini Yerdan anglab olish oson emas, chunki ikki yulduz orasidagi masofa ularning bizdan uzoqligidan ancha kichik va shuning uchun yulduzlarni teleskoplar bilan ajrata olmaymiz. Shuning uchun, ma'lum bir yulduzning qo'sh tizim ekanligini aniqlash uchun uning intensivligi yoki spektridagi o'zgarishlarni kuzatish uchun fotometriya (photometry) yoki spektrometriyadan (spectrometry) foydalanishimiz kerak.

Qo'sh yulduzlarning fotometriyasi (Photometry of Binary Stars)

Agar biz ikki yulduzning harakat tekisligida aniq joylashgan bo'lsak, unda bir yulduz ma'lum vaqtlarda ikkinchi yulduzni to'sib (occult) qo'yadi va butun tizimning intensivligi (intensity) bizning kuzatish nuqtamizdan vaqt bo'yicha o'zgaradi. Bunday qo'sh tizimlar tutiluvchi qo'sh yulduzlar (ecliptic binaries) deb ataladi.

1. Ikki yulduz ularning umumiy massa markazi atrofida aylana orbitalar (circular orbits) bo'ylab o'zgarmas burchak tezligi ω bilan harakatlanmoqda va biz tizimning harakat tekisligida aniq joylashganmiz deb faraz qiling. Shuningdek, yulduzlarning sirt haroratlari (surface temperatures) T_1 va T_2 ($T_1 > T_2$), mos keladigan radiuslari (radii) esa R_1 va R_2 ($R_1 > R_2$) deb faraz qiling. Yerda o'lchangan yorug'likning umumiy intensivligi (total intensity) vaqt funksiyasi sifatida 1-rasmda (Rasm (8).png) ko'rsatilgan. Ehtiyotkorlik bilan o'lchashlar shuni ko'rsatadiki, tushayotgan yorug'lik intensivligining minimal qiymatlari mos ravishda ikkala yulduzdan olingan maksimal intensivlik I_0 ning 90 va 63 foizini tashkil qiladi ($I_0 = 4.8 \times 10^{-9}\,\mathrm{W/m^2}$). 1-rasmdagi vertikal o'q I/I_0 nisbatini, gorizontal o'q esa kunlarda belgilangan vaqtni ko'rsatadi.

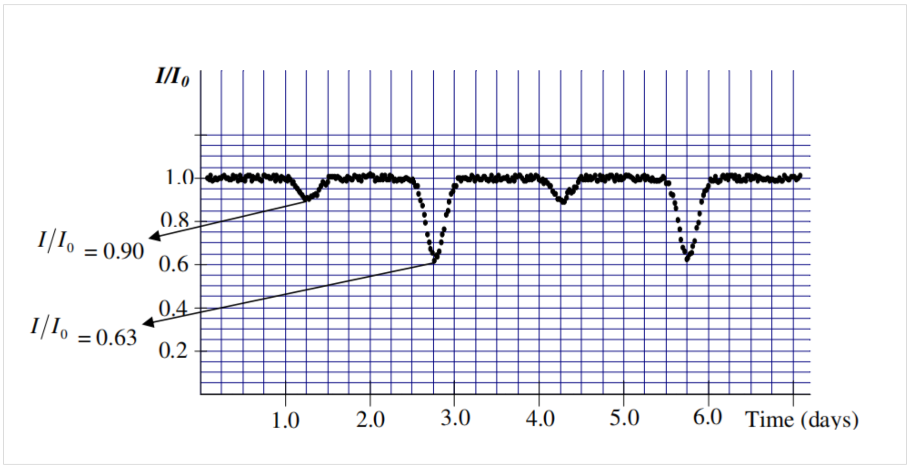


Figure 2: 1-rasm (Rasm (8).png): Qo'sh yulduz tizimidan olingan nisbiy intensivlikning vaqtga bog'liqligi. Vertikal o'q $I_0 = 4.8 \times 10^{-9}\,\mathrm{W/m^2}$ ga masshtablangan. Vaqt kunlarda berilgan.

1.1 Orbital harakatning davrini (period) toping. Javobingizni soniyalarda ikkita muhim raqamgacha bering. Tizimning burchak chastotasi (angular frequency) rad/sekundda qanday? (0.8 ball)

Yaxshi yaqinlashishda, yulduzdan olinadigan nurlanish radiusi yulduz radiusiga teng bo'lgan tekis diskdan (flat disc) bir xil qora jism nurlanishidir (uniform black body radiation). Shuning uchun, yulduzdan olinadigan quvvat (power) AT^4 ga proporsional, bu yerda A – diskning maydoni, T – yulduzning sirt harorati (surface temperature).

1.2 1-rasmdagi diagrammadan foydalanib, T_1/T_2 va R_1/R_2 nisbatlarini toping. (1.6 ball)

Qo'sh tizimlarning spektrometriyasi (Spectrometry of Binary Systems)

Ushbu bo'limda biz qo'sh yulduz tizimining eksperimental spektrometrik ma'lumotlaridan foydalanib, uning astronomik xususiyatlarini hisoblaymiz.

Atomlar o'zlariga xos to'lqin uzunliklarida nurlanishni yutadi yoki chiqaradi. Shuning uchun, yulduzning kuzatilgan spektri (observed spectrum) uning atmosferasidagi atomlar tufayli yutilish chiziqlarini (absorption lines) o'z ichiga oladi. Natriy (Sodium) o'ziga xos sariq chiziqli spektrga (D chizig'i) ega, to'lqin uzunligi 5895.9 Å (10 Å = 1 nm). Biz oldingi bo'limdagi qo'sh yulduz tizimi uchun atomik natriyning ushbu to'lqin uzunligidagi yutilish spektrini tekshiramiz. Qo'sh yulduzdan oladigan yorug'lik spektri Doppler siljishiga (Doppler-shifted) uchraydi, chunki yulduzlar bizga nisbatan harakatlanmoqda. Har bir yulduz har xil tezlikka ega. Shunga ko'ra, har bir yulduz uchun yutilish to'lqin uzunligi har xil miqdorga siljiydi. Doppler siljishini kuzatish uchun juda aniq to'lqin uzunlik o'lchovlari talab qilinadi, chunki yulduzlarning tezligi yorug'lik tezligidan ancha kichik. Biz ko'rib chiqayotgan qo'sh yulduz tizimining massa markazining tezligi yulduzlarning orbital tezliklaridan ancha kichik. Shuning uchun barcha Doppler siljishlarini yulduzlarning orbital tezligiga bog'lash mumkin. 1-jadvalda (Table 1) biz kuzatgan qo'sh yulduz tizimidagi yulduzlarning o'lchangan spektri ko'rsatilgan.

Table 1: 1-jadval (Table 1): Qo'sh yulduz tizimining natriy D chizig'i uchun yutilish spektri.

t/kun	0.3	0.6	0.9	1.2	1.5	1.8	2.1	2.4
λ_1 (Å)	5897.5	5897.7	5897.2	5896.2	5895.1	5894.3	5894.1	5894.6
λ_2 (Å)	5893.1	5892.8	5893.7	5896.2	5897.3	5898.7	5899.0	5898.1
t/kun	2.7	3.0	3.3	3.6	3.9	4.2	4.5	4.8
λ_1 (Å)	5895.6	5896.7	5897.3	5897.7	5897.2	5896.2	5895.0	5894.3
λ_2 (Å)	5896.4	5894.5	5893.1	5892.8	5893.7	5896.2	5897.4	5898.7

Table 2: *

(Eslatma: Ushbu jadval ma'lumotlarining grafigini chizish shart emas.)

2. 1-jadvaldan foydalanib,

   2.1 v_1 va v_2 har bir yulduzning orbital tezligi (orbital velocity) bo'lsin. v_1 va v_2 ni toping. Yorug'lik tezligi $c = 3.0 \times 10^8$ m/s. Relyativistik effektlarni e'tiborsiz qoldiring. (1.8 ball)

   2.2 Yulduzlarning massa nisbatini (mass ratio) m_1/m_2 toping. (0.7 ball)

   2.3 r_1 va r_2 har bir yulduzning massa markazidan masofalari (distances of each star from their center of mass) bo'lsin. r_1 va r_2 ni toping. (0.8 ball)

2.4 Yulduzlar orasidagi masofa r bo'lsin. r ni toping. (0.2 ball)

3. Yulduzlar orasida faqat tortishish kuchi (gravitational force) ta'sir qiladi.

3.1 Har bir yulduzning massasini bitta muhim raqamgacha toping. (1.2 ball) Universal tortishish doimiysi (universal gravitational constant) $G = 6.7 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$.

4. Yulduzlarning umumiy xarakteristikalari (General Characteristics of Stars)

Ko'pchilik yulduzlar energiyani bir xil mexanizm orqali hosil qiladi. Shu sababli, ularning massasi M va yorqinligi (luminosity) L (yulduzning umumiy nurlanish quvvati) o'rtasida empirik munosabat mavjud. Bu munosabat $L/L_{\text{quyosh}} = (M/M_{\text{quyosh}})^{\alpha}$ ko'rinishida yozilishi mumkin. Bu yerda $M_{\text{quyosh}} = 2.0 \times 10^{30}\,\text{kg}$ – Quyosh massasi, $L_{\text{quyosh}} = 3.9 \times 10^{26}\,\text{W}$ – Quyosh yorqinligi. Ushbu munosabat log-log diagrammasida 2-rasmda (Rasm (10).png) ko'rsatilgan.

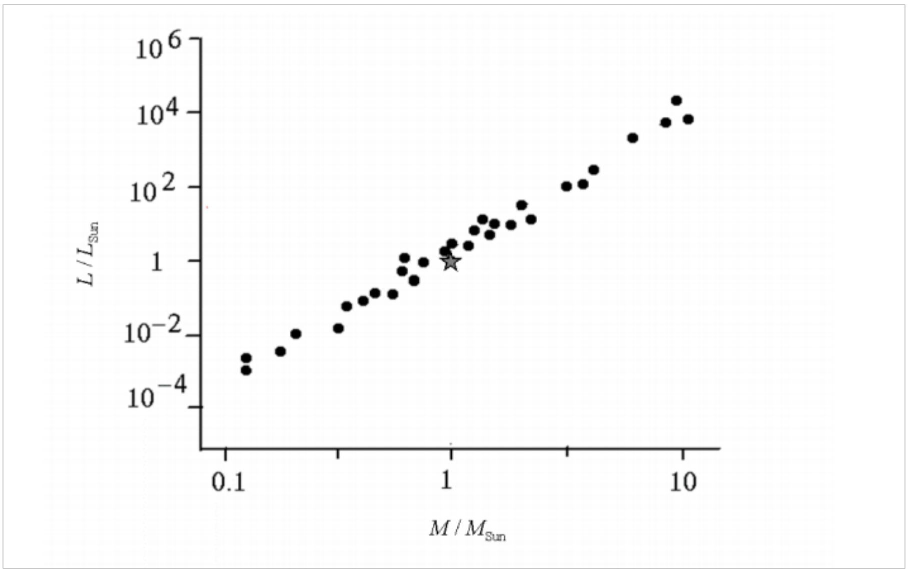


Figure 3: 2-rasm (Rasm (10).png): Yulduz yorqinligining uning massasiga bog'liqligi darajali qonun bo'yicha o'zgaradi. Diagramma log-log. Yulduz belgisi massasi $2.0 \times 10^{30}\,\text{kg}$ va yorqinligi $3.9 \times 10^{26}\,\text{W}$ bo'lgan Quyoshni ifodalaydi.

4.1 α ni bitta muhim raqamgacha toping. (0.6 ball)

4.2 Oldingi bo'limlarda o'rganilgan qo'sh yulduz tizimidagi yulduzlarning yorqinliklari L_1 va L_2 bo'lsin. L_1 va L_2 ni toping. (0.6 ball)

4.3 Yulduz tizimining bizdan masofasi d yorug'lik yilida (light years) qancha? Masofani topish uchun 1-rasmdagi diagrammadan foydalanishingiz mumkin. Bir yorug'lik yili – yorug'likning bir yilda bosib o'tadigan masofasi. (0.9 ball)

4.4 Yulduzlar orasidagi maksimal burchak masofasi (maximum angular distance) θ bizning kuzatish nuqtamizdan qancha? (0.4 ball)

4.5 Ushbu ikki yulduzni ajrata oladigan optik teleskopning eng kichik diafragma o'lchami (smallest aperture size) D qancha? (0.4 ball)

Eksperimental qurilmaning tavsifi (Description of the Apparatus)

1-rasmda (Rasm (11).png) sizning stolingizda o'rnatilgan qurilmaning umumiy ko'rinishini ko'rishingiz mumkin. Ushbu asbob spektroskop (spectroscope) bo'lib, oddiy spektrometr (spectrometer) vazifasini bajarish uchun detektor bilan jihozlangan.

Qurilmani sozlashni boshlash uchun avval qutining oq qopqog'ini (white cover) yuqoriga ko'tarishingiz kerak (1-rasm). Qopqoq asbob asosining bir tomoniga osilgan. Detektor uchun qorong'u muhit yaratish maqsadida, spektrlarni o'lchash vaqtida qopqoq dastlabki holatiga qaytarilishi va mahkam yopiq holda saqlanishi kerak. Quvvat simida halogen chiroqni (halogen lamp) yoqadigan va o'chiradigan kalit mavjud. Asbobni tekislash uchun to'rtta vint (screw) bor (kattalashtirilgan ko'rinishini 1-rasmning o'ng ichki qismida ko'rishingiz mumkin).

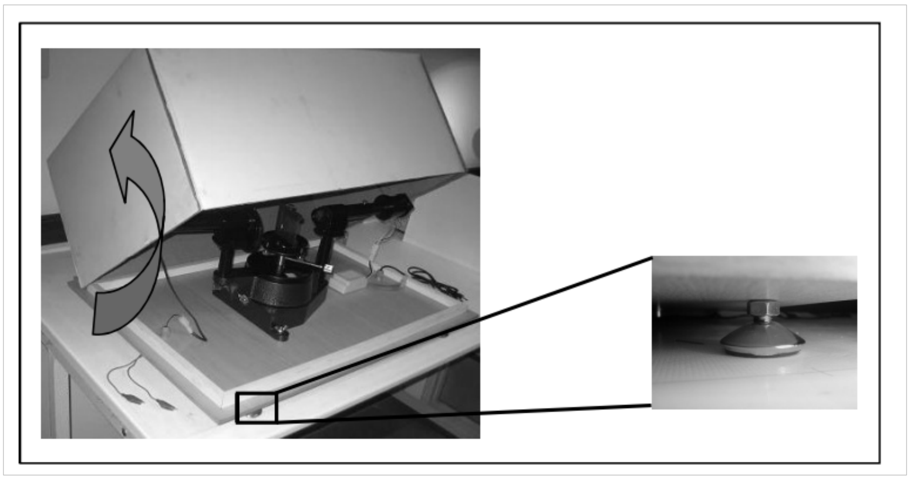


Figure 4: 1-rasm (Rasm (11).png): Eksperiment qurilmasi. O'ng ichki qismda tekislash vintlaridan biri kattalashtirilgan.

Diqqat 1: Halogen chiroq va uning ushlagichiga tegishdan saqlaning, chunki chiroq yoqilgandan keyin ular issiq bo'ladi!

Diqqat 2: Adapter va uning ulanishlariga aralashmang. Qurilmaga quvvat 220 V rozetkalar orqali beriladi!

Qurilmaning ustki ko'rinishi (top view) 2-rasmda (Rasm (12).png) ko'rsatilgan. Tafsilotlar rasmda keltirilgan.

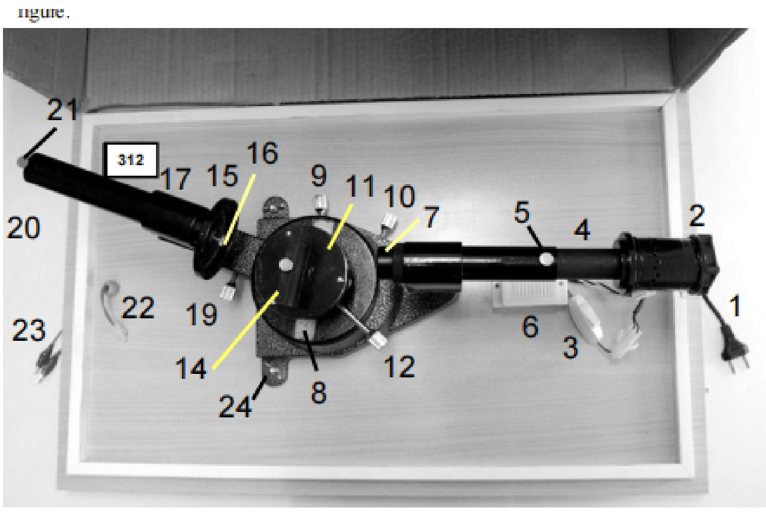


Figure 2.

1. Power cord
2. Halogen lamp and its cooling fan
3. On/Off switch
4. Arm of adjustable length
5. Adjusting screw
6. Adaptor: 220V – less than 12 V
7. Lens
8. Vernier
9. Vernier's lock
10. Fine adjustment screw for the vernier
11. Grating's stage
12. Grating's stage's fixing screw
13. Adjustment screw for leveling the grating's stage (shown in Fig. 4)
14. Grating holder
15. Sample holder
16. Fixing and adjusting screw for the sample and glass holder (Fig. 6)
17. Rotatable arm
18. Rotatable arm's lock (Fig.4)
19. Fine adjustment for the rotatable arm
20. Detector position
21. Fixing screw for the detector
22. Connecting socket for the detector
23. Connection to the multimeter
24. Fixing screw to the base

The number mentioned on the top-left corner, is the **apparatus number**.

Figure 5: 2-rasm (Rasm (12).png): Qurilmaning ustki ko'rinishi.

2-rasmdagi raqamli belgilar:

(a) Quvvat simi (Power cord)

(b) Halogen chiroq va uning sovutish ventilyatori (Halogen lamp and its cooling fan)

(c) Yoqish/o'chirish kaliti (On/Off switch)

(d) Uzunligi sozlanadigan qo'l (Arm of adjustable length)

(e) Sozlash vinti (Adjusting screw)

(f) Adapter: 220V – 12V dan kam (Adaptor: 220V – less than 12V)

(g) Linza (Lens)

(h) Nonius (Vernier)

(i) Noniusni mahkamlash vinti (Vernier's lock)

(j) Nonius uchun nozik sozlash vinti (Fine adjustment screw for the vernier)

(k) Panjara (grating) stoli (Grating's stage)

(l) Panjara stolini mahkamlash vinti (Grating's stage's fixing screw)

(m) Panjara stolini tekislash uchun sozlash vinti (4-rasmda ko'rsatilgan) (Adjustment screw for leveling the grating's stage)

(n) Panjara ushlagichi (Grating holder)

(o) Namuna ushlagichi (Sample holder)

(p) Namuna va shisha ushlagichni mahkamlash va sozlash vinti (6-rasm) (Fixing and adjusting screw for the sample and glass holder)

(q) Aylanadigan qo'l (Rotatable arm)

(r) Aylanadigan qo'lni mahkamlash vinti (4-rasm) (Rotatable arm's lock)

(s) Aylanadigan qo'l uchun nozik sozlash vinti (Fine adjustment for the rotatable arm)

(t) Detektor joylashuvi (Detector position)

(u) Detektorni mahkamlash vinti (Fixing screw for the detector)

(v) Detektor uchun ulash rozetkasi (Connecting socket for the detector)

(w) Multimetrga ulanish (Connection to the multimeter)

(x) Asosga mahkamlash vinti (Fixing screw to the base)

Yuqori chap burchakdagi raqam – asbob raqami.

Aylanadigan qo'lning qo'zg'almas qo'l yo'nalishi bilan hosil qilgan burchagi transportir (protractor) va nonius (vernier) yordamida o'lchanishi mumkin. Ushbu noniusning ajrata olish shkalasi (resolution scale) 30' (yoy daqiqasi). Ushbu asbob burchakni 5' aniqlikda o'lchay oladi.

Asbobga qo'shimcha ravishda siz 3-rasmda (Rasm (13).png) ko'rsatilgan qutichani topishingiz kerak, unda quyidagi elementlar mavjud:

(a) Detektor (detector) o'z ushlagichida;

(b) 600 chiziq/mm panjara (grating);

(c) Ramkaga o'rnatilgan namuna (sample) va shisha substrat (glass substrate).

Avval panjarani qutichasidan chiqarib, uning ramkasiga (panjara ushlagichi, 4-rasmga qarang) ehtiyotkorlik bilan joylashtirishingiz kerak.

DIQQAT: Panjaraning sirtiga tegish uning diffraktsiya samaradorligini (diffraction efficiency) jiddiy kamaytirishi yoki hatto uni shikastlashi mumkin!

Panjarani o'z holatida vertikal turishi uchun uchta sozlash vinti mavjud (4-rasm).

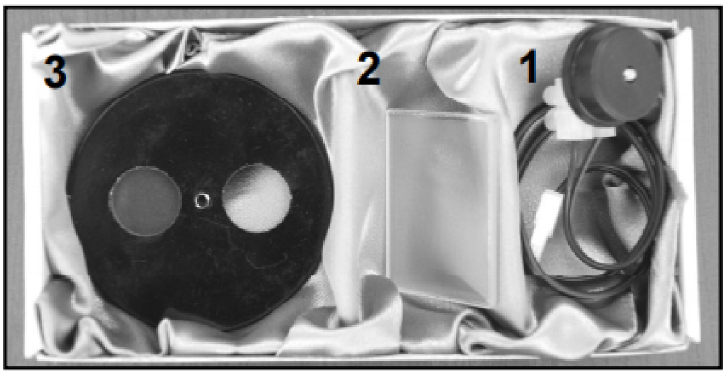


Figure 6: 3-rasm (Rasm (13).png): Shisha va namuna ushlagichi, difraksion panjara va fotorezistorni o'z ichiga olgan kichik quti.

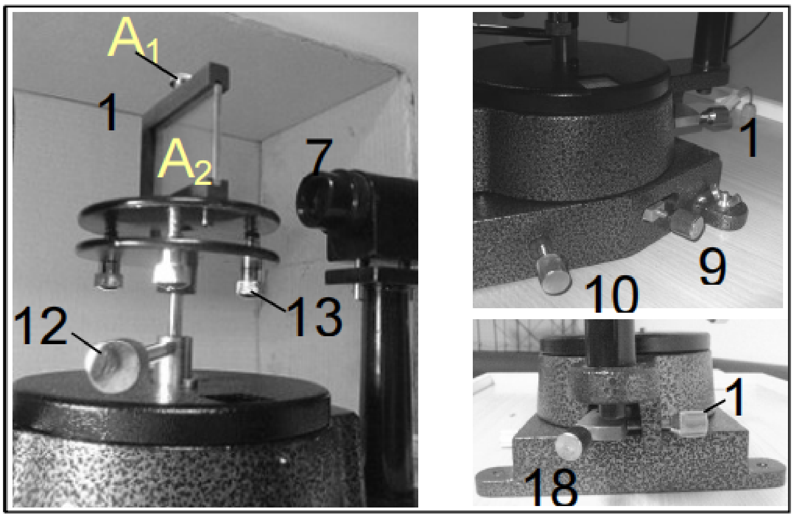


Figure 7: 4-rasm (Rasm (14).png): Qurilmaning mahkamlash, sozlash vintlari. A: Panjarani mahkamlash vinti; A: Panjara. 7, 9, 10, 12-14, 18 va 19 raqamlari 2-rasmda tushuntirilgan.

Detektor aylanadigan qo'lning oxirida o'z joyiga mahkam o'rnatilishi kerak (5-rasm – Rasm (15).png):

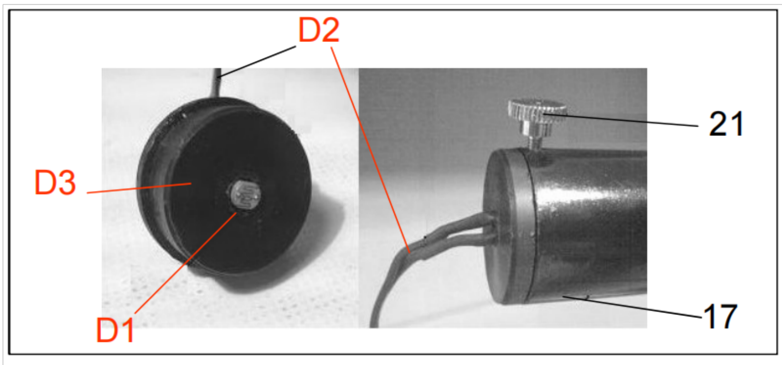


Figure 8: 5-rasm (Rasm (15).png): Detektor va uning ushlagichi. D1: Fotorezistor; D2: Ulovchi sim; D3: Detektor ushlagichi. 17 va 21 raqamlari 2-rasmda tushuntirilgan.

Namuna va shisha substrat ramkaga (ushlagichga) mahkamlangan (6c-rasm – Rasm (16).png), bu ramka mahkamlash vinti (6a-rasm, 16-band) yordamida asbobga biriktiriladi. Ushbu ramka aylanadigan bo'lib, namuna yoki shisha substratni kirish teshigi oldiga qo'yish mumkin, buning uchun ramkani mahkamlash vinti atrofida aylantirish kerak (6a-rasm).

Multimetr (The Multimeter)

Fotorezistor (photoresistor) tomonidan aniqlangan signalni qayd qilish uchun foydalaniladigan multimetr 7-rasmda (Rasm (17).png) ko'rsatilgan. Ushbu multimetr 200 M gacha o'lchay oladi. Qizil va qora prob simlari 7-rasmda ko'rsatilganidek asbobga ulanishi kerak. Yoqish/o'chirish tugmasi multimetrning chap tomonida joylashgan (7-rasm, M1 bandi).

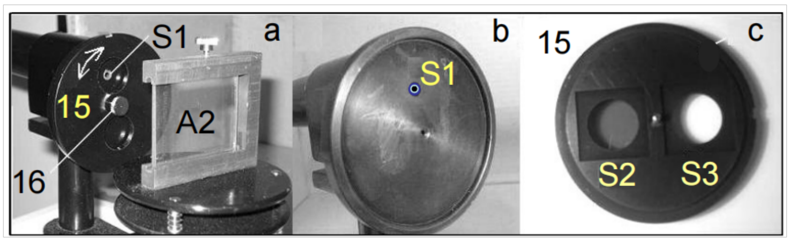


Figure 9: 6-rasm (Rasm (16).png): Namuna va shisha ushlagichi. S1: Kirish teshigi; S2: Namuna; S3: Shisha substrat. 15 va 16 raqamlari 2-rasmda tushuntirilgan.

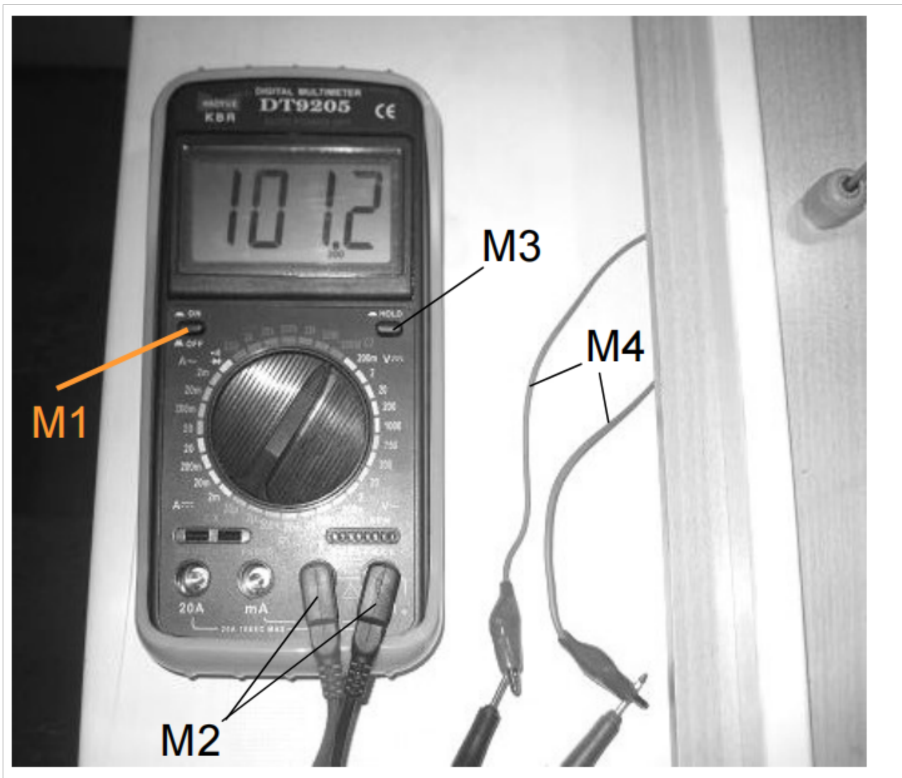


Figure 10: 7-rasm (Rasm (17).png): Fotorezistor qarshiligini o'lchash uchun multimetr. M1: yoqish/o'chirish kaliti; M2: prob simlari; M3: Hold tugmasi; M4: asbobga ulanishlar.

Eslatma: Multimetr avtomatik o'chirish xususiyatiga ega. Avtomatik o'chirish holatida, yoqish/o'chirish tugmachasini (M1) ketma-ket ikki marta bosishingiz kerak.

- Hold tugmasi tajriba davomida faol bo'lmasligi kerak.

Eksperimental masala: Yarim o'tkazgich yupqa plyonkalarining energiya zona oralig'ini aniqlash

(Determination of energy band gap of semiconductor thin films)

I. Kirish (Introduction)

Yarim o'tkazgichlar (semiconductors) taxminan o'tkazgichlar va izolyatorlar o'rtasida elektron xususiyatlarga ega bo'lgan materiallar sifatida tavsiflanishi mumkin. Yarim o'tkazgichlarning elektron xususiyatlarini tushunish uchun, fotoefekt (photoelectric effect) taniqli hodisa sifatida boshlash mumkin. Fotoefekt – bu kvant elektron hodisa bo'lib, unda fotoelektronlar (photoelectrons) elektromagnit nurlanishdan (ya'ni fotonlardan) yetarli energiyani yutish orqali materiyadan chiqariladi. Metallardan yorug'lik nurlanishi bilan elektronni chiqarish uchun zarur bo'lgan minimal energiya "ish chiqishi" (work function) deb ataladi. Shunday qilib, faqat ma'lum bir chegara chastotadan yuqori bo'lgan, ya'ni energiyasi $h\nu$ (h – Plank doimiysi) materialning ish chiqishidan katta bo'lgan fotonlar fotoelektronlarni chiqarib tashlashga qodir.

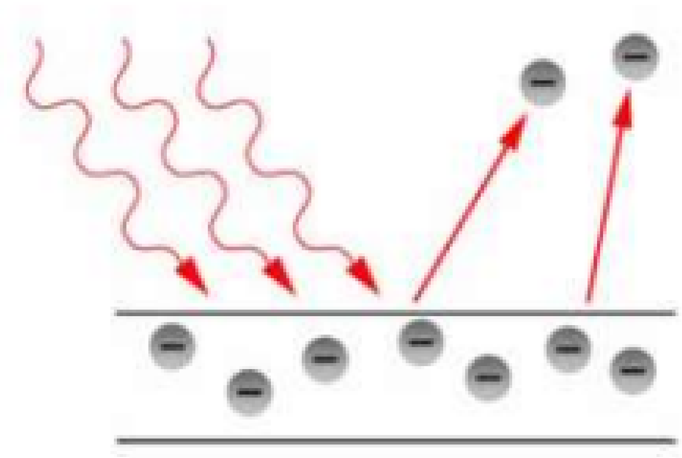


Figure 11: 1-rasm (Rasm (18).png): Metall plastinkadan fotoelektron emissiyasining tasviri: tushayotgan foton materialning ish chiqishidan katta energiyaga ega bo'lishi kerak.

Aslida, fotoeffekt jarayonidagi ish chiqishi tushunchasi yarim o'tkazgich materialining energiya zona oralig'i (energy band gap) tushunchasiga o'xshaydi. Qattiq jism fizikasida, zona oralig'i E_g izolyatorlar va yarim o'tkazgichlarning valent zonasi (valence band) tepasi va o'tkazuvchanlik zonasi (conduction band) tubi o'rtasidagi energiya farqidir. Valent zonasi elektronlar bilan to'liq to'ldirilgan, o'tkazuvchanlik zonasi esa bo'sh, ammo elektronlar etarli energiyaga ega bo'lsalar (kamida zona oralig'i energiyasiga teng) valent zonasidan o'tkazuvchanlik zonasiga o'tishlari mumkin. Yarim o'tkazgichning o'tkazuvchanligi (conductivity) uning energiya zona oralig'iga kuchli bog'liq.

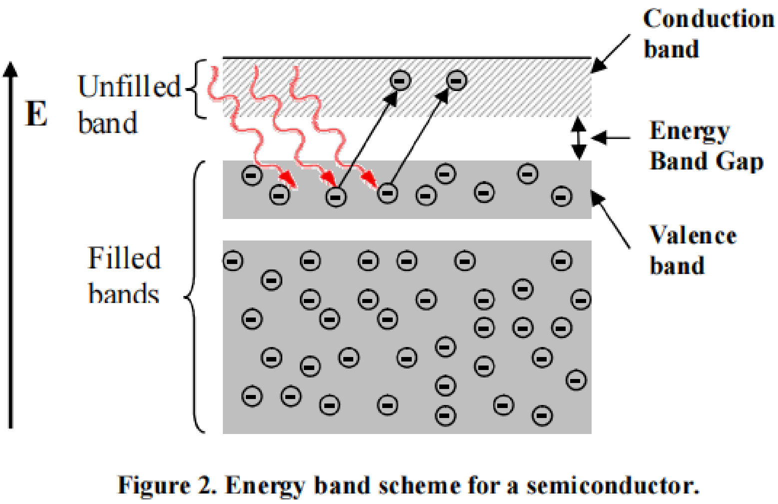


Figure 12: 2-rasm (Rasm (19).png): Yarim o'tkazgich uchun energiya zonasi sxemasi.

Zona muhandisligi (Band gap engineering)

Bu ma'lum yarim o'tkazgich qotishmalarining tarkibini nazorat qilish orqali materialning zona oralig'ini boshqarish yoki o'zgartirish jarayonidir. So'nggi paytlarda yarim o'tkazgichning nanostrukturasini o'zgartirish orqali uning zona oralig'ini manipulyatsiya qilish mumkinligi ko'rsatildi.

Ushbu tajribada biz optik usul yordamida temir oksidi (Fe_2O_3) nano-zarrachalar zanjirlarini o'z ichiga olgan yupqa plyonkali yarim o'tkazgichning energiya zona oralig'ini olishimiz kerak. Zona oralig'ini o'lchash uchun biz shaffof plyonkaning optik yutilish xususiyatlarini (optical absorption properties) uning optik o'tkazuvchanlik spektri (transmission spectrum) yordamida o'rganamiz. Taxminan aytganda, yutilish spektri (absorption spectrum) tushayotgan fotonlarning energiyasi energiya zona oralig'iga teng bo'lganda keskin o'sishni ko'rsatadi.

II. Eksperimental qurilma (Experimental Setup)

Sizning stolingizda quyidagi narsalarni topasiz:

(a) Tarkibida halogen chiroqli (halogen lamp) spektrometr (spectrometer) bo'lgan katta oq quti.

(b) Tarkibida namuna (sample), shisha substrat (glass substrate), namuna ushlagichi (sample-holder), panjara (grating) va fotorezistor (photoresistor) bo'lgan kichik quti.

(c) Multimetr (multimeter).

(d) Kalkulyator (calculator).

(e) Chizg'ich (ruler).

(f) Markazida teshik ochilgan karta (card with a hole punched in its center).

(g) Bir qator bo'sh yorliqlar (blank labels).

Spektrometr aniqlik (precision) 5' bo'lgan goniometrni (goniometer) o'z ichiga oladi. Halogen chiroq nurlanish manbai (source of radiation) vazifasini bajaradi va spektrometrning qo'zg'almas qo'liga (fixed arm) o'rnatilgan (batafsil ma'lumot uchun ilova qilingan "Qurilmaning tavsifi" ga qarang).

Kichik qutida quyidagi narsalar mavjud:

(a) Ikki oynali namuna ushlagichi (sample-holder with two windows): bir oynaga Fe_2O_3 plyonkasi qoplangan shisha substrat, ikkinchisiga esa qoplanmagan shisha substrat o'rnatilgan.

(b) Yorug'lik detektori (light detector) vazifasini bajaruvchi o'z ushlagichiga o'rnatilgan fotorezistor.

(c) Shaffof difraksion panjara (transparent diffraction grating) (600 chiziq/mm).

Eslatma: Kichik qutidagi har qanday komponentning sirtiga tegishdan saqlaning!

Qurilmaning sxematik diagrammasi 3-rasmda (Rasm (20).png) ko'rsatilgan:

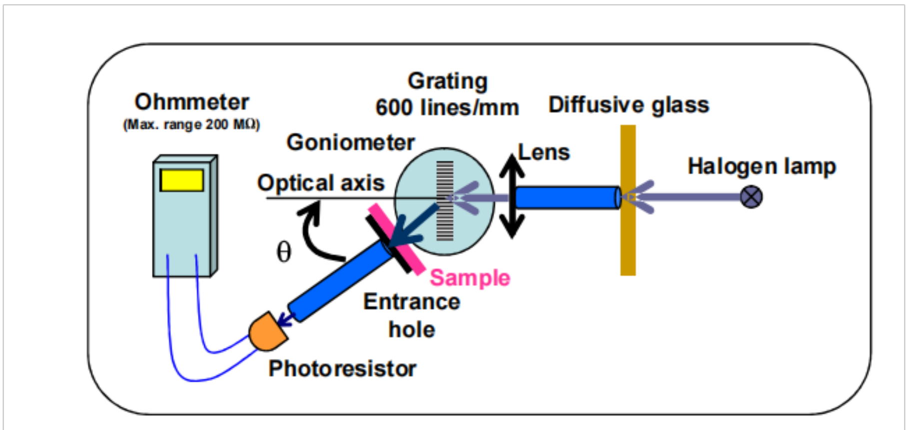


Figure 3. Schematic diagram of the experimental setup.

Figure 13: 3-rasm (Rasm (20).png): Eksperimental qurilmaning sxematik diagrammasi.

III. Usullar (Methods)

Har bir to'lqin uzunligida plyonkaning o'tkazuvchanligini (transmission) olish uchun quyidagi formuladan foydalanish mumkin:

$$T_{\text{film}}(\lambda) = \frac{I_{\text{film}}(\lambda)}{I_{\text{glass}}(\lambda)} \tag{1}$$

bu yerda I_{film} va I_{glass} mos ravishda qoplangan shisha substratdan o'tgan yorug'lik intensivligi (intensity) va qoplanmagan shisha slayddan o'tgan yorug'lik intensivligi. I qiymatini fotorezistor kabi yorug'lik detektori yordamida o'lchash mumkin. Fotorezistorda tushayotgan yorug'lik intensivligi oshganda elektr qarshilik (electrical resistance) kamayadi. Bu erda I qiymati quyidagi munosabatdan aniqlanishi mumkin:

$$I(\lambda) = C(\lambda)R^{-1} \tag{2}$$

bu yerda R – fotorezistorning elektr qarshiligi, C – λ ga bog'liq koeffitsiyent.

Spektrometrdagi shaffof panjara (transparent grating) har xil to'lqin uzunlikdagi yorug'likni har xil burchaklarga diffraktsiyalaydi (diffracts). Shuning uchun, T ning λ ga bog'liq o'zgarishlarini o'rganish uchun fotorezistorning burchagini (θ') optik o'qga (optical axis) (panjaraga tushayotgan yorug'lik nurining yo'nalishi sifatida aniqlanadi) nisbatan o'zgartirish kifoya, 4-rasmda (Rasm (21).png) ko'rsatilganidek.

Difraksion panjaraning asosiy tenglamasidan:

$$n\lambda = d[\sin(\theta' - \theta_0) + \sin\theta_0] \tag{3}$$

ma'lum bir λ ga mos keladigan θ' burchakni olish mumkin: n – diffraktsiya tartibini (order of diffraction) ifodalovchi butun son, d – panjara davri (period of the grating), θ_0 – panjara yuzasiga normal vektorning optik o'q bilan hosil qilgan burchagi (4-rasmga qarang). (Ushbu tajribada biz panjarani optik o'qga perpendikulyar joylashtirishga harakat qilamiz, ya'ni $\theta_0 = 0$, ammo buni mukammal aniqlik bilan amalga oshirish mumkin emasligi sababli, ushbu sozlash bilan bog'liq xatolik 1-e topshirig'ida o'lchanadi.)

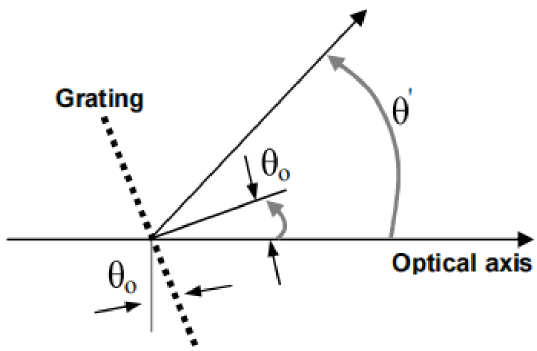


Figure 4. Definition of the angles involved in Equation 3.

Figure 14: 4-rasm (Rasm (21).png): (3) tenglamadagi burchaklarning ta'rifi.

Eksperimental ravishda, zona oralig'i energiyasidan biroz katta foton energiyalari uchun quyidagi munosabat o'rinli ekanligi ko'rsatilgan:

$$\alpha h\nu = A(h\nu - E_g)^{\eta} \tag{4}$$

bu yerda α – plyonkaning yutilish koeffitsiyenti (absorption coefficient), A – plyonka materialiga bog'liq konstanta, η – plyonka materiali va strukturasining yutilish mexanizmi (absorption mechanism) bilan aniqlanadigan konstanta. O'tkazuvchanlik (transmission) α ga taniqli yutilish munosabati orqali bog'liq:

$$T_{\text{film}} = \exp(-\alpha t) \tag{5}$$

bu yerda t – plyonkaning qalinligi (thickness).

IV. Topshiriqlar (Tasks)

0. Sizning asbobingiz va namuna qutingiz (namuna ushlagichini o'z ichiga olgan kichik quti) raqamlar bilan belgilangan. Javoblar varag'ida tegishli qutilarga Asbob raqami (Apparatus number) va Namuna raqamini (Sample number) yozing.

1. Sozlashlar va o'lchovlar (Adjustments and Measurements)

1-a Nonius shkalasini (vernier scale) tekshiring va maksimal aniqlikni ($\Delta\theta$) hisobot qiling. (0.1 ball)

Eslatma: Kattalashtiruvchi ko'zoynaklar (Magnifying glasses) so'rov bo'yicha mavjud.

1-qadam: Tajribani boshlash uchun halogen chiroqni isitish uchun yoqing. Tajriba davomida chiroqni o'chirmaslik yaxshiroq. Halogen chiroq tajriba davomida qizib ketganligi sababli, ehtiyot bo'ling va unga tegmang.

Chiroqni linzadan imkon qadar uzoqroqqa qo'ying, bu sizga parallel yorug'lik nurini (parallel light beam) beradi.

Fotorezistordan foydalanmasdan goniometrning taxminiy nol-sozlashini (rough zero-adjustment) qilmoqchimiz. Vint 18 (qo'l ostida) bilan aylanadigan qo'lni bo'shating va aylanadigan qo'lni optik o'q bilan vizual ravishda tekislang. Endi vint 18 bilan aylanadigan qo'lni mahkamlang. Vint 9 bilan noniusni bo'shating va stolni nonius shkalasida 0 ga aylantiring. Endi vint 9 bilan noniusni mahkamlang va noniusning nozik sozlash vintidan (vint 10) foydalanib, nonius shkalasining nolini sozlang. Panjarani uning ushlagichiga joylashtiring. Difraksion panjara optik o'qga taxminan perpendikulyar bo'lguncha panjara stolini aylantiring. Markazida teshik bo'lgan kartani yorug'lik manbai oldiga qo'ying va teshikni shunday joylashtiringki, yorug'lik nurlari panjaraga tushsin. Panjarani aylantiring, aks ettirilgan yorug'lik nuqtasi teshikka tushsin. Shunda aks ettirilgan yorug'lik nuri tushayotgan nur bilan ustma-ust tushadi. Endi vint 12 ni mahkamlash orqali panjara stolini mahkamlang.

1-b Teshik va panjara orasidagi masofani o'lchab, ushbu sozlashning aniqligini ($\Delta\theta$) hisoblang. (0.3 ball)

Endi aylanadigan qo'lni aylantirib, ko'rinadigan yorug'likning (qizildan ko'kgacha) birinchi tartibli diffraktsiyasi kuzatiladigan burchaklar oralig'ini aniqlang va hisobot qiling. (0.2 ball)

2-qadam: Endi fotorezistorni aylanadigan qo'lning oxiriga o'rnating. Tizimni optik jihatdan tekislash uchun fotorezistordan foydalanib, vint 18 ni bo'shating va aylanadigan qo'lni fotorezistor minimal qarshilikni ko'rsatguncha biroz burang. Aniq joylashish uchun vint 18 ni mahkamlang va aylanadigan qo'lning nozik sozlash vintidan foydalaning. Noniusning nozik sozlash vinti yordamida nonius shkalasining nolini sozlang.

1-c O'lchangan minimal qarshilik qiymatini ($R^{(0)}_{\min}$) hisobot qiling. (0.1 ball)

Sizning nol-sozlashingiz endi aniqroq, ushbu yangi sozlashning aniqligini ($\Delta\varphi_0$) hisobot qiling. (0.1 ball)

Eslatma: $\Delta\varphi_0$ – bu tekislashdagi xatolik, ya'ni aylanadigan qo'l va optik o'qning noto'g'ri tekislanganligi o'lchovidir.

- **Maslahat**: Ushbu topshiriqdan so'ng noniusning mahkamlash vintlarini mahkamlang. Bundan tashqari, fotorezistor ushlagichining vintini mahkamlang va uni tajriba davomida olib tashlamang.

3-qadam: Aylanadigan qo'lni birinchi tartibli diffraktsiya sohasiga olib boring. Fotorezistor qarshiligi minimal (yorug'lik intensivligi maksimal) bo'lgan burchakni toping. Balanslash vintlari yordamida panjara stolining egilishini biroz o'zgartirib, undan ham pastroq qarshilik qiymatiga erishishingiz mumkin.

1-c Kuzatilgan minimal qarshilik qiymatini ($R^{(1)}_{\text{min}}$) tegishli qutiga yozing. (0.1 ball)

Endi nol sozlash uchun panjaraning perpendikulyarligini qayta tekshirish kerak. Buning uchun 1-qadamdagi aks ettirish-moslik usulidan foydalanishingiz kerak.

Muhim: Bundan keyin tajribani qorong'uda o'tkazing (qopqoqni yoping).

O'lchovlar: Namuna ushlagichini aylanadigan qo'lga vidalang. O'lchovlarni boshlashdan oldin yarim o'tkazgich plyonkangizning (namuna) tashqi ko'rinishini tekshiring. Namunani aylanadigan qo'ldagi kirish teshigi S_1 oldiga shunday joylashtiringki, namunaning bir tekis qoplangan qismi teshikni qoplasin. Har safar namunaning bir xil qismi bilan ishlayotganingizga ishonch hosil qilish uchun namuna ushlagichi va aylanadigan qo'liga bo'sh yorliqlar bilan tegishli belgilar qo'ying.

Diqqat: Yuqori qarshilik o'lchovlarida fotorezistorning relaksatsiyasiga vaqt berish kerak, shuning uchun ushbu diapazondagi har bir o'lchov uchun o'lchovni yozib olishdan oldin 3-4 daqiqa kuting.

1-d Qoplanmagan shisha substrat va yarim o'tkazgich qatlami bilan qoplangan shisha substrat uchun fotorezistor qarshiligini θ burchak (goniometr tomonidan fotorezistor va belgilangan optik o'q orasidagi burchak uchun o'qilgan qiymat) funksiyasi sifatida o'lchang. So'ngra 1d-jadvalni to'ldiring. 1b-qadamda topilgan diapazonda kamida 20 ta ma'lumot nuqtasiga ega bo'lishingiz kerak. O'lchovlaringizni ohmmetringizning tegishli diapazonidan foydalanib bajaring. (2.0 ball)

Har bir ma'lumot nuqtasi bilan bog'liq xatolikni hisobga oling. (1.0 ball)

4-qadam (Step 4)

Hozirgacha olingan aniqlik hali cheklangan, chunki aylanadigan qo'lni optik o'q bilan tekislash va/yoki panjarani optik o'qga perpendikulyar joylashtirishni 100% aniqlik bilan amalga oshirish mumkin emas. Shuning uchun biz hali ham o'lchangan o'tkazuvchanlikning optik o'qning ikkala tomonidagi assimetriyasini (bu panjara yuzasiga normalning optik o'qdan og'ishi natijasida kelib chiqadi) topishimiz kerak.

Ushbu assimetriyani o'lchash uchun quyidagi bosqichlarni bajaring:

1-e Avval T_{nw} ni $\theta = -20°$ da o'lchang. So'ngra $+20°$ atrofidagi boshqa burchaklarda T_{nw} qiymatlarini oling. 1e-jadvalni to'ldiring (siz 1d-jadvalda olingan qiymatlardan foydalanishingiz mumkin). (0.6 ball)

T_{nw} ni θ ga nisbatan grafigini chizing va vizual ravishda egri chiziq chizing. (0.6 ball)

Sizning egri chiziqingizda T_{nw} qiymati $\theta = -20°$ da o'lchagan T_{nw} ga teng bo'lgan γ burchakni toping (γ dan $\theta = -20°$). Ushbu burchakning $+20°$ bilan farqini δ deb belgilang, boshqacha aytganda:

$$\delta = \gamma - 20°$$

1-e δ qiymatini belgilangan qutiga yozing. (0.2 ball)

Keyin birinchi tartibli diffraktsiya uchun (3) tenglamani quyidagicha soddalashtirish mumkin:

$$\lambda = d \sin(\theta - \delta/2) \tag{7}$$

bu yerda θ – goniometrda o'qilgan burchak.

2. Hisob-kitoblar (Calculations)

2-a (7) tenglamadan foydalanib, $\Delta\lambda$ ni boshqa parametrlarning xatoliklari orqali ifodalang (δ aniq deb faraz qiling va u bilan bog'liq xatolik yo'q). Shuningdek, (1), (2) va (5) tenglamalardan foydalanib, ΔT_{nw} ni R va ΔR orqali ifodalang. (0.6 ball)

2-b Birinchi tartibli diffraktsiya sohasida $\Delta\lambda$ qiymatlarining oralig'ini hisobot qiling. (0.3 ball)

2-c 1-topshiriqdagi o'lchangan parametrlarga asoslanib, har bir θ uchun 2c-jadvalni to'ldiring. Eslatma: to'lqin uzunligi (7) tenglama yordamida hisoblanishi kerak. (2.4 ball)

2-d R_{nw} va R^{-}_{nw} ni to'lqin uzunligi funksiyasi sifatida bir xil diagrammada birgalikda grafigini chizing. Eslatma: (2) tenglama asosida R_{nw} va R^{-}_{nw} ning harakati mos ravishda I_{glass} va I_{film} ning harakati haqida asosli ko'rsatma berishi mumkin. (1.5 ball)

2d-jadvalda R_{nw} va R^{-}_{nw} minimal qiymatlariga erishadigan to'lqin uzunliklarini hisobot qiling. (0.4 ball)

Yarim o'tkazgich qatlami (namuna) uchun T_{film} ni to'lqin uzunligi funksiyasi sifatida grafigini chizing. Bu miqdor plyonka o'tkazuvchanligining to'lqin uzunligi bo'yicha o'zgarishini ham ifodalaydi.

3. Ma'lumotlarni tahlil qilish (Data analysis)

(4) tenglamada $\eta = 1/2$ va $A = 0.071\,(\text{eV})^{1/2}/\text{nm}$ ni qo'yib, E_g va t (mos ravishda eV va nm birliklarida) qiymatlarini topish mumkin. Bunga $x - y$ koordinata tizimida mos diagramma chizish va ushbu tenglamani qanoatlantiradigan sohada ekstrapolyatsiya qilish orqali erishiladi.

$x = h\nu$ va $y = (\alpha t h\nu)^2$ deb faraz qilib va 1-topshiriqdagi o'lchovlaringizdan foydalanib, 530 nm va undan yuqori to'lqin uzunliklari uchun 3a-jadvalni to'ldiring. Natijalaringizni (x va y) bitta ma'lumot nuqtasidagi xatolik bahosiga asoslanib, to'g'ri muhim raqamlar bilan ifodalang. Eslatma: $h\nu$ eV birligida, to'lqin uzunligi esa nm birligida hisoblanishi kerak. Jadvalning yuqori qatoridagi qavslar orasiga har bir o'zgaruvchining birligini yozing.

3-a y ni x ga nisbatan grafigini chizing.

y parametri plyonkaning yutilishiga mos kelishiga e'tibor bering. Taxminan 530 nm atrofidagi chiziqli sohadagi nuqtalarga to'g'ri chiziq moslang (fit a line).

Chiziqni moslagan ma'lumot nuqtalarining eng kichik va eng katta x-koordinatalari qiymatlarini hisobot qilib, (4) tenglama bajariladigan sohani aniqlang.

3-b Ushbu chiziqning qiyaligini m deb ataymiz va plyonka qalinligi (t) va uning xatoligi (Δt) uchun m va A orqali ifoda toping (A ni xatosiz deb hisoblang). (1.0 ball)

Tahlilingiz uchun zarur bo'lgan ba'zi foydali fizik konstantalar:

- Yorug'lik tezligi: $c = 3.00 \times 10^8\,\text{m/s}$
- Plank doimiysi: $h = 6.63 \times 10^{-34}\,\text{J} \cdot \text{s}$
- Elektron zaryadi: $e = 1.60 \times 10^{-19}\,\text{C}$

Nazariy masala yechimlari (Theory Question Solutions)

1.1

So'ralgan kattaliklarning o'lchovligini olish uchun har qanday mantiqiy tenglamadan foydalanish mumkin.

- I) Plank munosabati $h\nu = E \implies [h][\nu] = [E] \implies [h] = [E][\nu]^{-1} = ML^2T^{-1}$ (0.2 ball)
- II) $[c] = LT^{-1}$ (0.2 ball)
- III) $F = \frac{Gmm}{r^2} \implies [G] = [F][r^2][m]^{-2} = M^{-1}L^3T^{-2}$ (0.2 ball)
- IV) $E = k_B\theta \implies [k_B] = [\theta]^{-1}[E] = ML^2T^{-2}K^{-1}$ (0.2 ball)

1.2

Stefan–Boltsman qonunidan foydalanib,

$$\frac{\text{Quvvat}}{\text{Maydon}} = \sigma\theta^4$$

yoki har qanday ekvivalent munosabatdan: (0.3 ball)

$$[\sigma]K^4 = [E]L^2T^{-1} \implies [\sigma] = MT^{-3}K^{-4} \qquad \text{(0.2 ball)}$$

1.3

Stefan–Boltsman doimiysi son koeffitsiyentgacha quyidagiga teng:

$$\sigma = h^\alpha c^\beta G^\gamma k_B^\delta$$

bu yerda $\alpha, \beta, \gamma, \delta$ olchovli tahlil bilan aniqlanishi mumkin. Haqiqatan,

$$[\sigma] = [h]^\alpha[c]^\beta[G]^\gamma[k_B]^\delta$$

masalan, $[\sigma] = MT^{-3}K^{-4}$.

$$\begin{aligned} MT^{-3}K^{-4} &= (ML^2T^{-1})^\alpha(LT^{-1})^\beta(M^{-1}L^3T^{-2})^\gamma(ML^2T^{-2}K^{-1})^\delta \\ &= M^{\alpha-\gamma+\delta}L^{2\alpha+\beta+3\gamma+2\delta}T^{-\alpha-\beta-2\gamma-2\delta}K^{-\delta} \qquad \text{(0.2 ball)} \end{aligned}$$

Yuqoridagi tenglik bajarilishi uchun:

$$\Rightarrow \begin{cases} \alpha - \gamma + \delta = 1, \\ 2\alpha + \beta + 3\gamma + 2\delta = 0, \\ -\alpha - \beta - 2\gamma - 2\delta = -3, \\ -\delta = -4, \end{cases} \quad \text{(Har biri 0.1 ball)} \implies \begin{cases} \alpha = -3, \\ \beta = -2, \\ \gamma = 0, \\ \delta = 4. \end{cases} \quad \text{(Har biri 0.1 ball)}$$

$\implies \sigma = \frac{k_B^4}{c^2h^3}$.

2.1

A (hodisa ufqi maydoni) m ga nisbatan klassik relyativistik gravitatsiya nazariyasidan (masalan, Umumiy nisbiylik nazariyasi) hisoblanishi kerak, bu c (maxsus nisbiylik nazariyasiga xos) va G (gravitatsiyaga xos) ning kombinatsiyasidir. Ayniqsa, u Plank doimiysi h (kvant mexanik hodisalarga xos) ga bog'liq emas.

$$A = G^{\alpha}c^{\beta}m^{\gamma}$$

O'lchovli tahlildan foydalanib:

$$[A] = [G]^{\alpha}[c]^{\beta}[m]^{\gamma} \Rightarrow L^2 = (M^{-1}L^3T^{-2})^{\alpha}(LT^{-1})^{\beta}M^{\gamma} = M^{-\alpha+\gamma}L^{3\alpha+\beta}T^{-2\alpha-\beta}$$

(0.2 ball)

Tenglik bajarilishi uchun:

$$\Rightarrow \begin{cases} -\alpha+\gamma=0, \\ 3\alpha+\beta=2, \\ -2\alpha-\beta=0, \end{cases} \quad \text{(Har biri 0.1 ball)} \Longrightarrow \begin{cases} \alpha=2, \\ \beta=-4, \\ \gamma=2, \end{cases} \quad \text{(Har biri 0.1 ball)}$$

$\Longrightarrow A = \frac{m^2G^2}{c^4}$.

2.2

Entropiyaning ta'rifidan $dS = \frac{dQ}{\theta}$:

$$[S] = [E][\theta]^{-1} = ML^2T^{-2}K^{-1} \qquad \text{(0.2 ball)}$$

2.3

$\eta = S/A$ ekanligini e'tiborga olib:

$$[\eta] = [S][A]^{-1} = MT^{-2}K^{-1}$$

$$[\eta] = [G]^{\alpha}[h]^{\beta}[c]^{\gamma}[k_B]^{\delta} = M^{-\alpha+\beta+\delta}L^{3\alpha+2\beta+\gamma+2\delta}T^{-2\alpha-\beta-\gamma-2\delta}K^{-\delta} \qquad \text{(0.2 ball)}$$

Yuqoridagi sxema bo'yicha:

$$\Rightarrow \begin{cases} -\alpha+\beta+\delta=1, \\ 3\alpha+2\beta+\gamma+2\delta=0, \\ -2\alpha-\beta-\gamma-2\delta=-2, \\ \delta=1, \end{cases} \quad \text{(Har biri 0.1 ball)} \Longrightarrow \begin{cases} \alpha=-1, \\ \beta=-1, \\ \gamma=3, \\ \delta=1, \end{cases} \quad \text{(Har biri 0.1 ball)}$$

Shunday qilib,

$$\eta = \frac{c^3k_B}{Gh} \qquad \text{(0.1 ball)}$$

3.1

... (davomi keyingi sahifalarda)

3.1

Termodinamikaning birinchi qonuni (first law of thermodynamics): $dE = dQ + dW$. Farazga ko'ra, $dW = 0$. Entropiya (entropy) ta'rifi $dS = \frac{dQ}{\theta}$ dan foydalanib:

$$dE = \theta_H dS + 0, \quad (0.2) + (0.1) \text{ (} dW = 0 \text{ qilish uchun).}$$

Quyidagilardan foydalanib:

$$S = \frac{Gk_B m^2}{c\hbar}, \quad [0.1] \text{ (} S \text{ uchun)}$$

$$E = mc^2,$$

hosil qilamiz:

$$\theta_H = \frac{dE}{dS} = \left(\frac{dS}{dE}\right)^{-1} = c^2 \left(\frac{dS}{dm}\right)^{-1} \quad (0.2)$$

Shuning uchun:

$$\theta_H = \frac{1}{2}\frac{c^3\hbar}{Gk_B m} \quad (0.1) + (0.1) \text{ (koeffitsiyent uchun)}$$

3.2

Stefan–Boltsman qonuni (Stefan-Boltzmann's law) birlik sirt uchun energiya nurlanish tezligini (rate of energy radiation) beradi. $E = mc^2$ ekanligini e'tiborga olib:

$$\begin{aligned} dE/dt &= -\sigma\theta_H^4 A, \quad (0.2) \\ \sigma &= \frac{k_B^4}{c^2\hbar^3}, \\ A &= \frac{m^2G^2}{c^4}, \\ E &= mc^2. \end{aligned}$$

$$\Rightarrow \frac{dm}{dt} = -\frac{1}{16}\frac{c^4\hbar}{G^2m^2} \quad (0.1) \text{ (soddalashtirish uchun)}+(0.2) \text{ (minus ishorasi uchun)} \quad (0.3)$$

3.3

Integrallash orqali:

$$\frac{dm}{dt} = -\frac{1}{16}\frac{c^4\hbar}{G^2m^2}, \quad \Rightarrow \int m^2 dm = -\int \frac{c^4\hbar}{16G^2}dt \quad (0.3)$$

$$\Rightarrow m^3(t) - m^3(0) = -\frac{3c^4\hbar}{16G^2}t, \quad (0.2)+(0.2) \text{ (Integrallash va to'g'ri chegara qiymatlari)} \quad (0.4)$$

$t = t^*$ da qora tuynuk to'liq bug'lanadi (evaporates completely):

$$m(t^*) = 0 \quad (0.1) \quad \Rightarrow t^* = \frac{16G^2}{3c^4\hbar}m^3 \quad (0.2) + (0.1) \text{ (koeffitsiyent uchun)} \quad (0.5)$$

3.4

C_V – E ning θ ga nisbatan o'zgarishini o'lchaydi (heat capacity):

$$C_V = \frac{dE}{d\theta},$$
$$E = mc^2,$$
$$\theta = \frac{c^3\hbar}{2Gk_B m}$$

$$C_V = \frac{2Gk_B}{c\hbar}m^2. \quad (0.1) + (0.1) \text{ (koeffitsiyent uchun)}$$

4.1

Yana Stefan–Boltsman qonuni qora tuynukning birlik sirt maydonidan energiya yo'qotish tezligini (rate of energy loss) beradi. Qora tuynukning fon nurlanishidan (background radiation) energiya olish tezligini olish uchun shunga o'xshash munosabatdan foydalanish mumkin. Buni asoslash uchun, termal muvozanatda (thermal equilibrium) energiyaning umumiy o'zgarishi nolga teng ekanligini e'tiborga oling. Qora jism nurlanishi (blackbody radiation) Stefan–Boltsman qonuni bilan beriladi. Shuning uchun energiya olish tezligi ham xuddi shu formula bilan beriladi.

(0.1) + (0.4) (mos ravishda birinchi va ikkinchi hadlar uchun)

$$\frac{dE}{dt} = -\sigma\theta^4 A + \sigma\theta_B^4 A$$
$$E = mc^2,$$
$$\frac{dm}{dt} = -\frac{\hbar c^4}{16G^2}\frac{1}{m^2} + \frac{G^2}{c^8\hbar^3}(k_B\theta_B)^4 m^2$$

4.2

$\frac{dm}{dt} = 0$ deb qo'yib:

$$-\frac{\hbar c^4}{16G^2}\frac{1}{m^2} + \frac{G^2}{c^8\hbar^3}(k_B\theta_B)^4 m^2 = 0 \tag{0.2}$$

va natijada:

$$m^* = \frac{c^3\hbar}{2Gk_B\theta_B} \tag{0.2}$$

4.3

$$\theta_B = \frac{c^3\hbar}{2Gk_B m^*} \quad \Rightarrow \quad \frac{dm}{dt} = -\frac{\hbar c^4}{16G^2}\frac{1}{m^2}\left(1 - \frac{m^4}{m^{*4}}\right) \tag{0.2}$$

4.4

4.2 dagi yechimdan foydalanib,

$$m^* = \frac{c^3\hbar}{2Gk_B\theta_B} \quad \text{va 3.1 dan,} \quad \theta^* = \frac{c^3\hbar}{2Gk_B m^*} = \theta_B \tag{0.2}$$

Yana bir usul: m^* termal muvozanatga mos keladi. Shuning uchun $m = m^*$ uchun qora tuynuk harorati θ_B ga teng. Yoki $\frac{dE}{dt} = -\sigma(\theta^{*4} - \theta_B^4)A = 0$ dan $\theta^* = \theta_B$ ni olish mumkin.

4.5

4.3 dagi yechimni ko'rib chiqib, tizim muvozanatdan uzoqlashishini tekshiramiz. (0.6 ball)

$$\frac{dm}{dt} = -\frac{\hbar c^4}{16G^2}\frac{1}{m^2}\left(1 - \frac{m^4}{m^{*4}}\right) \Rightarrow \begin{cases} m > m^* & \Rightarrow \frac{dm}{dt} > 0 \\ m < m^* & \Rightarrow \frac{dm}{dt} < 0 \end{cases}$$

Bu muvozanat beqaror (unstable) ekanligini ko'rsatadi.

Masala “Orange” (Kondensator va akselerometr)

1.1

Bir plastinka uchun Gauss qonunidan (Gauss’s law) elektr maydonini (electric field) olamiz:

$$E = \frac{\sigma}{\varepsilon_0} \tag{0.2}$$

Zaryadi Q va maydoni A bo’lgan plastinka uchun sirt zaryad zichligi (surface charge density):

$$\sigma = \frac{Q}{A} \tag{0.2}$$

Elektr maydoni ikkita ekvivalent parallel plastinka tomonidan hosil qilinadi. Shuning uchun har bir plastinkaning maydonga qo’shgan hissasi $\frac{1}{2}E$ ga teng. Kuch (force) elektr maydoni zaryadga ko’paytirilganiga teng:

$$\text{Force} = \frac{1}{2}EQ = \frac{Q^2}{2\varepsilon_0 A} + (0.2)\ (\ \frac{1}{2}\ \text{koeffitsiyenti} + \text{yakuniy natija}) \tag{0.2}$$

1.2

Prujina (spring) uchung Guk qonuni (Hook’s law):

$$F_m = -kx \tag{0.2}$$

1.1 da ikki plastinka orasidagi elektr kuchini topdik:

$$F_e = \frac{Q^2}{2\varepsilon_0 A}$$

Tizim barqaror. Muvozanat sharti (equilibrium condition):

$$F_m = F_e \tag{0.2}$$

$$\Rightarrow x = \frac{Q^2}{2\varepsilon_0 A k} \tag{0.2}$$

1.3

Elektr maydoni o’zgarmas, shuning uchun potensiallar farqi (potential difference) V:

$$V = E(d - x) \tag{0.2}$$

(Boshqa mantiqiy yondashuvlar ham qabul qilinadi. Masalan, sig’im ta’rifidan foydalanish mumkin.) Oldingi qismdan olingan elektr maydonini yuqoridagi tenglamaga qo’yib:

$$V = \frac{Qd}{\varepsilon_0 A}\left(1 - \frac{Q^2}{2\varepsilon_0 A k d}\right) \tag{0.2}$$

1.4

C zaryadning potensiallar farqiga nisbati sifatida aniqlanadi:

$$C = \frac{Q}{V} \tag{0.1}$$

1.3 javobidan foydalanib:

$$\frac{C}{C_0} = \left(1 - \frac{Q^2}{2\varepsilon_0 Akd}\right)^{-1} \tag{0.2}$$

1.5

Bizda prujinadan keladigan mexanik energiya (mechanical energy):

$$U_m = \frac{1}{2}kx^2, \tag{0.2}$$

va kondensatorda saqlanadigan elektr energiyasi (electrical energy):

$$U_E = \frac{Q^2}{2C}. \tag{0.2}$$

Shuning uchun tizimda saqlanayotgan umumiy energiya (total energy):

$$U = \frac{Q^2 d}{2\varepsilon_0 A}\left(1 - \frac{Q^2}{4\varepsilon_0 Akd}\right) \tag{0.2}$$

2.1

Berilgan x qiymati uchun har bir kondensatordagi zaryad (charge on each capacitor):

$$Q_1 = VC_1 = \frac{\varepsilon_0 AV}{d-x}, \tag{0.2}$$

$$Q_2 = VC_2 = \frac{\varepsilon_0 AV}{d+x}. \tag{0.2}$$

2.2

Ikkita kondensator mavjud. Har bir kondensator uchun 1.1 dagi javobdan foydalanib:

$$F_1 = \frac{Q_1^2}{2\varepsilon_0 A}, \quad F_2 = \frac{Q_2^2}{2\varepsilon_0 A}.$$

Bu ikki kuch qarama-qarshi yo'nalgan, shuning uchun aniq elektr kuchi (net electric force):

$$F_E = F_1 - F_2, \Rightarrow F_E = \frac{\varepsilon_0 AV^2}{2}\left(\frac{1}{(d-x)^2} - \frac{1}{(d+x)^2}\right) \tag{0.2}$$

2.3

2.2 javobida x^2 tartibli hadlarni e'tiborsiz qoldirib (ignoring terms of order x^2):

$$F_E = \frac{2\varepsilon_0 A V^2}{d^3} x \tag{0.2}$$

2.4

Ikkita prujina bir xil prujina konstantasi k bilan ketma-ket (series) joylashgan, shuning uchun mexanik kuch (mechanical force):

$$F_m = -2kx \quad \text{(chapdagi prujina uchun)?}$$

Asl matnda: "There are two springs placed in series with the same spring constant, k, then the mechanical force is" – davomi keltirilmagan. To'liq yechim uchun davomi kerak, ammo berilgan matn shu joyda tugaydi. Qolgan qism (2.4, 2.5 va boshqalar) ushbu fayllarda mavjud emas. Shuning uchun biz faqat mavjud matnni kiritamiz.

2.4 (davomi)

Ikkita prujina ketma-ket joylashgan, har biri bir xil prujina konstantasi k ga ega, shuning uchun mexanik kuch:

$$F_m = -2kx. \quad \text{(Koeffitsiyent (2) uchun 0.2 ball)}$$

Bu natijani 2.3 javobi bilan birlashtirib va bu ikki kuch qarama-qarshi yo'nalganligini e'tiborga olib:

$$F = F_m + F_E, \quad \Rightarrow \quad F = -2\left(k - \frac{\varepsilon_0 A V^2}{d^3}\right) x,$$

(Ikki kuchning qarama-qarshi ishoralari uchun 0.3 ball)

$$\Rightarrow \quad k_{\text{eff}} = 2\left(k - \frac{\varepsilon_0 A V^2}{d^3}\right) \quad \text{(0.2 ball)}$$

2.5

Nyutonning ikkinchi qonunidan foydalanib:

$$F = ma \quad \text{(0.2 ball)}$$

va 2.4 javobidan:

$$a = -\frac{2}{m}\left(k - \frac{\varepsilon_0 A V^2}{d^3}\right) x \quad \text{(0.2 ball)}$$

3.1

Kirxgof qonunlaridan (Kirchhoff's laws) boshlab, ikkita elektr zanjiri uchun:

$$\begin{cases} \frac{Q_s}{C_s} + V - \frac{Q_2}{C_2} = 0 \\ -\frac{Q_s}{C_s} + V - \frac{Q_1}{C_1} = 0 \end{cases}$$

eslatma: ishoralar tanlangan maxsus sxemaga bog'liq bo'lishi mumkin)

$$Q_2 - Q_1 + Q_s = 0$$

$V_s = \frac{Q_s}{C_s}$ ekanligini e'tiborga olib:

$$\Rightarrow V_s = V \frac{\frac{2\varepsilon_0 Ax}{d^2 - x^2}}{C_s + \frac{2\varepsilon_0 Ad}{d^2 - x^2}}.$$

(Yuqoridagi tenglamalarni yechish uchun 0.4 ball va yakuniy natija uchun 0.2 ball)

Eslatma: Talabalar $d^2 \gg x^2$ yaqinlashishidan foydalanib yuqoridagi munosabatni soddalashtirishlari mumkin. Bu bo'limda farq qilmaydi.

3.2

3.1 javobida x^2 tartibli hadlarni e'tiborsiz qoldirib:

$$V_s = V \frac{2\varepsilon_0 Ax}{d^2 C_s + 2\varepsilon_0 Ad} \quad \text{(0.2 ball)}$$

4.1

Elektr kuchining mexanik (prujina) kuchiga nisbati:

$$\frac{F_E}{F_m} = \frac{\varepsilon_0 AV^2}{kd^3}$$

Raqamli qiymatlarni qo'yib:

$$\frac{F_E}{F_m} = 7.6 \times 10^{-9}$$

(0.2 ball – kattalik tartibi uchun, 0.2 ball – ikki muhim raqam uchun, 0.2 ball – to'g'ri javob (7.6 yoki 7.5) uchun). Bu natijadan ko'rinib turibdiki, elektr kuchlarini prujina kuchlariga nisbatan e'tiborsiz qoldirish mumkin.

4.2

Oldingi bo'limda ko'rinib turganidek, harakatlanuvchi plastinkaga ta'sir qiluvchi yagona kuch prujinalardan keladi deb faraz qilish mumkin:

$$F = 2kx$$

(Muvozanat tushunchasi 0.2 ball). Demak, mexanik muvozanatda harakatlanuvchi plastinkaning siljishi:

$$x = \frac{ma}{2k}$$

Maksimal siljish bu miqdorning ikki baravariga teng, xuddi tortishish maydonidagi massa-prujina tizimida massa qo'yib yuborilgandagi kabi:

$$x_{\max} = 2x \quad (0.2 \text{ ball})$$

$$x_{\max} = \frac{ma}{k} \quad (0.2 \text{ ball})$$

4.3

Tezlanish $a = g$ bo'lganda (0.2 ball):

$$x_{\max} = \frac{mg}{k}$$

Bundan tashqari, 3.2 da olingan natijadan:

$$V_s = V \frac{2\varepsilon_0 A x_{\max}}{d^2 C_s + 2\varepsilon_0 A d}$$

Bu masalada berilgan qiymat – 0.15 V ga teng bo'lishi kerak.

$$\Rightarrow \quad C_s = \frac{2\varepsilon_0 A}{d}\left(\frac{V x_{\max}}{V_s d} - 1\right) \quad (0.2 \text{ ball})$$

$$\Rightarrow \quad C_s = 8.0 \times 10^{-11} \text{ F} \quad (0.2 \text{ ball})$$

4.4

Haydovchining boshi va rul orasidagi masofa ℓ bo'lsin. Uni taxminan $\ell = 0.4\,\text{m} - 1\,\text{m}$ deb baholash mumkin (0.2 ball). Tezlanish boshlangan paytda haydovchi boshining avtomobilga nisbatan nisbiy tezligi nolga teng:

$$\Delta v(t = 0) = 0, \quad (0.2 \text{ ball})$$

Keyin:

$$\ell = \frac{1}{2} g t_1^2 \quad \Rightarrow \quad t_1 = \sqrt{\frac{2\ell}{g}} \quad (0.2 \text{ ball})$$

$$t_1 = 0.3 - 0.5\,\text{s} \quad (0.2 \text{ ball})$$

4.5

t_2 vaqti garmonik ossillyator (harmonic oscillator) davrining yarmiga teng, shuning uchun:

$$t_2 = \frac{T}{2}, \quad (0.3 \text{ ball})$$

Garmonik ossillyatorning davri oddiygina:

$$T = 2\pi\sqrt{\frac{m}{2k}}, \quad (0.2 \text{ ball})$$

shuning uchun:

$$t_2 = 0.013\,\text{s}. \quad (0.2 \text{ ball})$$

$t_1 > t_2$ bo'lgani uchun, havo yostig'i vaqtida ishga tushadi (0.2 ball).

Masala “Pink” (Qosh yulduzlar)

1.1

Davr (period) = 3.0 kun = 2.6×10^5 s. (0.4 ball)

$$\text{Davr} = \frac{2\pi}{\omega} \quad \text{(0.2 ball)} \Rightarrow \omega = 2.4 \times 10^{-5}\,\text{rad/s.} \quad \text{(0.2 ball)}$$

1.2

1-diagrammadagi minimalarni nomlaymiz: $I_1/I_0 = \alpha = 0.90$ va $I_2/I_0 = \beta = 0.63$. U holda:

$$\frac{I_0}{I_1} = 1 + \left(\frac{R_2}{R_1}\right)^2 \left(\frac{T_2}{T_1}\right)^4 = \frac{1}{\alpha} \quad \text{(0.4 ball)}$$

$$\frac{I_2}{I_1} = 1 - \left(\frac{R_2}{R_1}\right)^2 \left(1 - \left(\frac{T_2}{T_1}\right)^4\right) = \frac{\beta}{\alpha} \quad \text{(0.4 ball)} \quad \text{(yoki ekvivalent munosabatlar)}$$

Yuqoridagilardan:

$$\frac{R_1}{R_2} = \sqrt{\frac{\alpha}{1-\beta}} \Rightarrow \frac{R_1}{R_2} = 1.6 \quad (0.2 + 0.2\ \text{ball})$$

$$\frac{T_1}{T_2} = \sqrt{\frac{1-\beta}{1-\alpha}} \Rightarrow \frac{T_1}{T_2} = 1.4 \quad (0.2 + 0.2\ \text{ball})$$

2.1

Doppler siljishi formulasi (Doppler-Shift formula):

$$\frac{\Delta\lambda}{\lambda_0} = \frac{v}{c} \quad \text{(yoki ekvivalent munosabat)} \quad \text{(0.4 ball)}$$

Maksimal va minimal to’lqin uzunliklari:

$$\lambda_{1,\text{max}} = 5897.7\,\text{Å}, \quad \lambda_{1,\text{min}} = 5894.1\,\text{Å}$$

$$\lambda_{2,\text{max}} = 5899.0\,\text{Å}, \quad \lambda_{2,\text{min}} = 5892.8\,\text{Å}$$

Maksimal va minimal to’lqin uzunliklari orasidagi farq:

$$\Delta\lambda_1 = 3.6\,\text{Å}, \quad \Delta\lambda_2 = 6.2\,\text{Å} \quad \text{(Hammasi 0.6 ball)}$$

Doppler munosabatidan foydalanib va siljish orbital tezlikning ikki baravariga teng ekanligini e’tiborga olib (ikki baravar koeffitsiyent 0.4 ball):

$$v_1 = c\frac{\Delta\lambda_1}{2\lambda_0} = 9.2 \times 10^4\,\text{m/s} \quad \text{(0.2 ball)}$$

$$v_2 = c\frac{\Delta\lambda_2}{2\lambda_0} = 1.6 \times 10^5\,\text{m/s} \quad \text{(0.2 ball)}$$

(Talaba markaziy chiziq va maksimal (yoki minimal) to’lqin uzunliklaridan foydalanishi mumkin. Baholash sxemasi Excel faylida berilgan.)

2.2

Massa markazi bizga nisbatan harakatlanmagani uchun (0.5 ball):

$$\frac{m_1}{m_2} = \frac{v_2}{v_1} = 1.7 \quad (0.2 \text{ ball})$$

2.3

$r_i = \frac{v_i}{\omega}$ yozib ($i = 1, 2$), (0.4 ball):

$$r_1 = 3.8 \times 10^9 \, \text{m}, \quad (0.2 \text{ ball})$$

$$r_2 = 6.5 \times 10^9 \, \text{m} \quad (0.2 \text{ ball})$$

2.4

Yulduzlar orasidagi masofa:

$$r = r_1 + r_2 = 1.0 \times 10^{10} \, \text{m} \quad (0.2 \text{ ball})$$

3.1

Gravitatsion kuch massaning markazdan qochma tezlanishga (centrifugal acceleration) ko'paytmasiga teng:

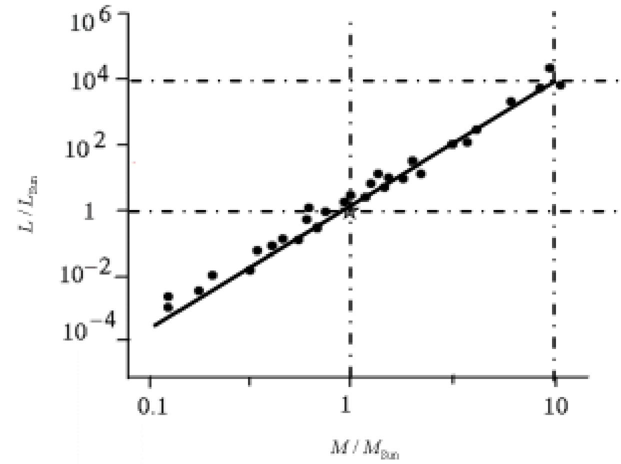
$$G\frac{m_1 m_2}{r^2} = m_1 \frac{v_1^2}{r_1} = m_2 \frac{v_2^2}{r_2} \quad (0.7 \text{ ball})$$

Shuning uchun:

$$\begin{cases} m_1 = \frac{r^2 v_2^2}{G r_2} \\ m_2 = \frac{r^2 v_1^2}{G r_1} \end{cases} \Rightarrow \begin{cases} m_1 = 6 \times 10^{30} \, \text{kg} \\ m_2 = 3 \times 10^{30} \, \text{kg} \end{cases} \quad (0.2 + 0.2 \text{ ball})$$

4.1

Diagrammadan ko'rinib turibdiki, bitta muhim raqam bilan $\alpha = 4$. (0.6 ball)



Oldingi bo'limda topganimizdek:

$$L_i = L_{\text{Quyosh}} \left(\frac{M_i}{M_{\text{Quyosh}}} \right)^4 \quad (0.2 \text{ ball})$$

Shunday qilib,

$$L_1 = 3 \times 10^{28}\,\text{Vt} \quad (0.2 \text{ ball})$$
$$L_2 = 4 \times 10^{27}\,\text{Vt} \quad (0.2 \text{ ball})$$

4.3

Tizimning umumiy quvvati radiusi d bo'lgan sfera yuzasiga tarqalib, I_0 ni hosil qiladi:

$$I_0 = \frac{L_1 + L_2}{4\pi d^2} \quad (0.5 \text{ ball})$$
$$\Rightarrow d = \sqrt{\frac{L_1 + L_2}{4\pi I_0}} = 1 \times 10^{18}\,\text{m} \quad (0.2 \text{ ball})$$
$$= 100\,\text{yorug'lik yili (ly)} \quad (0.2 \text{ ball})$$

4.4

$$\theta \equiv \tan\theta = \frac{r}{d} = 1 \times 10^{-8}\,\text{rad} \quad (0.2 + 0.2 \text{ ball})$$

4.5

Oddiy optik to'lqin uzunligi λ_0 dir. Noaniqlik munosabatidan foydalanib:

$$D = \frac{d\lambda_0}{r} \equiv 50\,\text{m} \quad (0.2 + 0.2 \text{ ball})$$

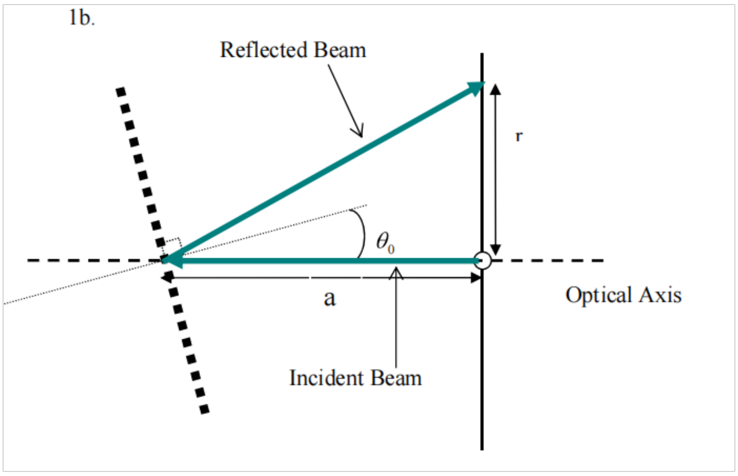
Eksperimental masala (davomi)

1-topshiriq (Task 1)

1a

$$\Delta\theta_{\text{nominal}} = 5' = 0.08°$$

1b



{Rasm: Aks ettirilgan nur va tushayotgan nur. Optik o'q.

Figure 15: Aks ettirilgan nurni tekislash sxemasi.

Agar "a" – karta va panjara orasidagi masofa, "r" – teshik va yorug'lik nuqtasi orasidagi masofa bo'lsa, u holda

$$\tan(2\theta_0) = \frac{r}{a}, \quad \text{Agar } \theta_0 < 1 \Rightarrow \theta_0 = \frac{r}{2a}$$

Xatolik:

$$\Delta\theta_0 = \sqrt{\left(\frac{\Delta r}{2a}\right)^2 + \left(\frac{r\Delta a}{2a^2}\right)^2}$$

Biz $\theta_0 = 0$ ni xohlaymiz, ya'ni $r = 0 \Rightarrow \Delta\theta_0 = \frac{\Delta r}{2a}$.
$\Delta r = 1\,\text{mm}$, $a = (70 \pm 1)\,\text{mm} \Rightarrow \Delta\theta_0 = \frac{1}{2\cdot 70}\,\text{rad} = 0.007\,\text{rad} = 0.4°$.

$$\boxed{\Delta\theta_0 = 0.4°}$$

Ko'rinadigan yorug'likning burchak oralig'i (degree):

$$13° \leq \theta \leq 26°$$

1c

$R_{\min}^{(0)}$	$(21.6 \pm 0.1)\,\mathrm{k\Omega}$
$\Delta\varphi_0$	$5' = 0.08°$
$R_{\min}^{(1)}$	$(192 \pm 1)\,\mathrm{k\Omega}$

$\Delta\varphi_0 = 5'$ chunki:

$$\theta = 5' \Rightarrow R = (21.9 \pm 0.1)\,\mathrm{k\Omega}, \qquad \theta = -5' \Rightarrow R = (21.9 \pm 0.1)\,\mathrm{k\Omega}.$$

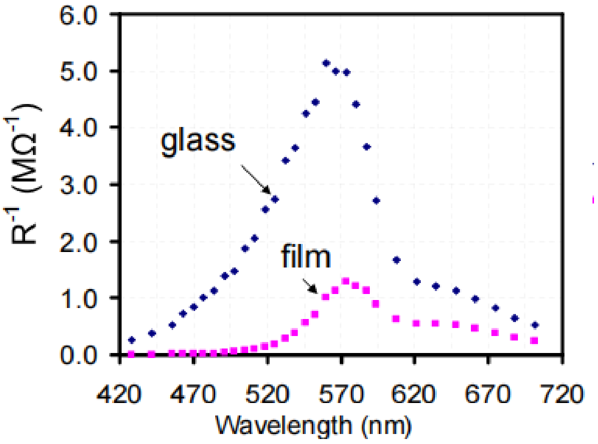
1d

O'lchangan parametrlar (Measured parameters): 1d-jadval.

θ (grad)	R_{glass} (MΩ)	$\Delta R_{\mathrm{glass}}$ (MΩ)	R_{film} (MΩ)	ΔR_{film} (MΩ)
15.00	3.77	0.03	183	3
15.50	2.58	0.02	132	2
16.00	1.88	0.01	87	1
16.50	1.19	0.01	51.5	0.5
17.00	0.89	0.01	33.4	0.3
17.50	0.68	0.01	19.4	0.1
18.00	0.486	0.005	10.4	0.1
18.50	0.365	0.005	5.40	0.03
19.00	0.274	0.003	2.66	0.02
19.50	0.225	0.002	1.42	0.01
20.00	0.200	0.002	0.880	0.005
20.50	0.227	0.002	0.822	0.005
21.00	0.368	0.003	1.123	0.007
21.50	0.600	0.005	1.61	0.01
22.00	0.775	0.005	1.85	0.01
22.50	0.83	0.01	1.87	0.01
23.00	0.88	0.01	1.93	0.02
23.50	1.01	0.01	2.14	0.02
24.00	1.21	0.01	2.58	0.02
24.50	1.54	0.01	3.27	0.02
25.00	1.91	0.01	4.13	0.02
16.25	1.38	0.01	66.5	0.5
16.75	1.00	0.01	40.0	0.3
17.25	0.72	0.01	23.4	0.2
17.75	0.535	0.005	12.8	0.1
18.25	0.391	0.003	6.83	0.05
18.75	0.293	0.003	3.46	0.02
19.25	0.235	0.003	1.76	0.01
19.75	0.195	0.002	0.988	0.005
20.25	0.201	0.002	0.776	0.005
20.75	0.273	0.003	0.89	0.01

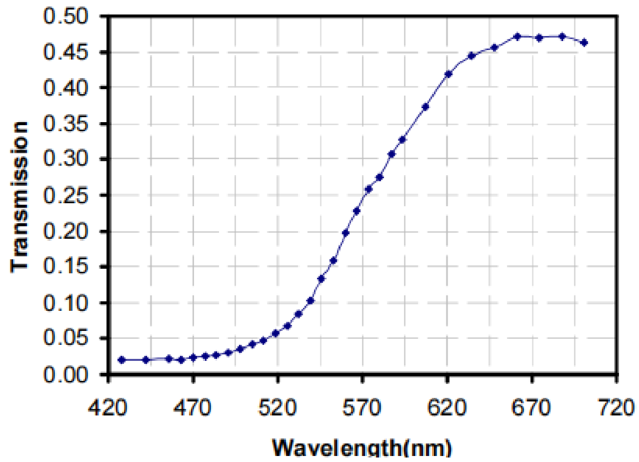
1e

Graphics



$\lambda_{max}(I_{glass})$	564±5 (nm)
$\lambda_{max}(I_{film})$	573±5 (nm)

2e. Graphics



$\theta = -20^\circ$ da: $R_{\text{glass}} = (132 \pm 2)\,\text{k}\Omega$, $R_{\text{film}} = (518 \pm 5)\,\text{k}\Omega$

θ	T_{film}	θ	T_{film}
$\theta = -20^\circ$	0.255	19.25	0.134
		19.50	0.158
		19.75	0.197
		20.00	0.227
		20.25	0.259
		20.50	0.276
		20.75	0.307

Grafik: T_{film} vs θ – egri chiziq. $T_{\text{film}}|_{\theta=-20^\circ}$ belgilangan.

Figure 16: Transmission burchakka bog'liqligi.

Grafikdan ko'ramiz: $T(\theta = 20.25°) = T(\theta = -20°)$.

$$\delta = 0.25° \pm 0.08°$$

2-topshiriq (Task 2)

2a

$$\lambda = d\sin\left(\theta - \frac{\delta}{2}\right) \Rightarrow \Delta\lambda = \lambda\sqrt{\left(\frac{\Delta d}{d}\right)^2 + \cot^2\left(\theta - \frac{\delta}{2}\right)\left(\Delta\theta^2 + \frac{\Delta\delta^2}{4}\right)} \approx d\cos(\theta)\left(\frac{0.1\pi}{180}\right)$$

bu yerda $\Delta\theta = \Delta\delta = 5' = 0.08^\circ$ va $d = \frac{1}{600}$ mm.

$$\Delta\lambda = 2.9\cos(\theta)\,(\mathrm{nm})$$

$$T_{\mathrm{film}} = \frac{R_{\mathrm{glass}}}{R_{\mathrm{film}}} \Rightarrow \Delta T = T_{\mathrm{film}}\sqrt{\left(\frac{\Delta R_{\mathrm{film}}}{R_{\mathrm{film}}}\right)^2 + \left(\frac{\Delta R_{\mathrm{glass}}}{R_{\mathrm{glass}}}\right)^2}$$

2b

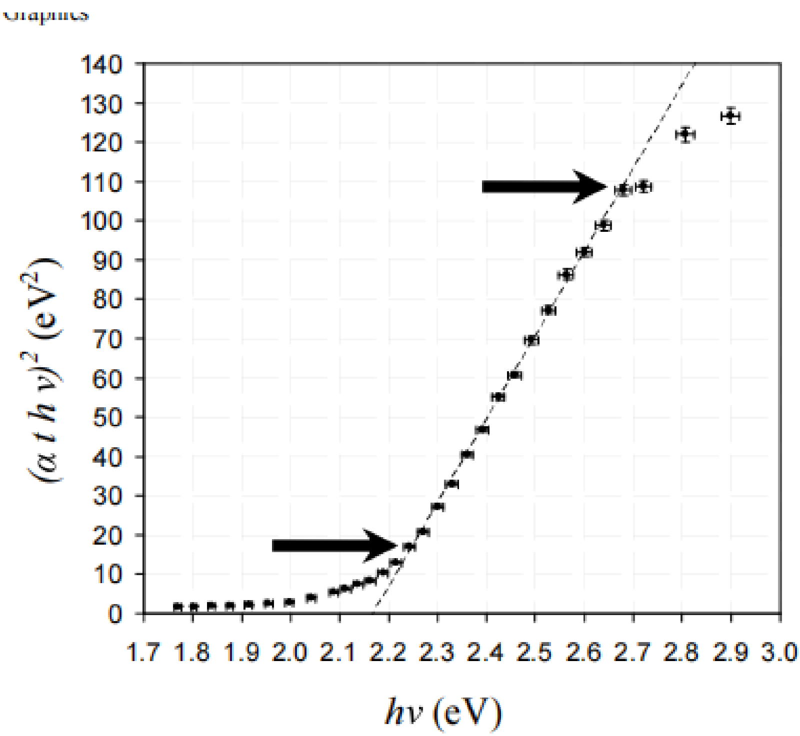
$13^\circ \leq \theta \leq 26^\circ$ oraliqda:

$$2.6 \leq \Delta\lambda \leq 2.8\,\mathrm{nm}$$

2c-jadval (Table 2c). O'lchangan parametrlar yordamida hisoblangan parametrlar

θ (grad)	λ (nm)	$I_g/C(\lambda)$ ($M\Omega^{-1}$)	$I_s/C(\lambda)$ ($M\Omega^{-1}$)	T_{film}	αt
15.0	428	0.265	0.00546	0.0206	3.88
15.5	442	0.388	0.00758	0.0195	3.94
16.0	456	0.532	0.0115	0.0216	3.83
16.25	463	0.725	0.0150	0.0208	3.88
16.5	470	0.840	0.0194	0.0231	3.77
16.75	477	1.00	0.0250	0.0250	3.69
17.0	484	1.12	0.0299	0.0266	3.63
17.25	491	1.39	0.0427	0.0308	3.48
17.5	498	1.47	0.0515	0.0351	3.35
17.75	505	1.87	0.0781	0.0418	3.17
18.0	512	2.06	0.096	0.0467	3.06
18.25	518	2.56	0.146	0.0572	2.86
18.5	525	2.74	0.185	0.0676	2.69
18.75	532	3.41	0.289	0.0847	2.47
19.0	539	3.65	0.376	0.103	2.27
19.25	546	4.26	0.568	0.134	2.01
19.5	553	4.44	0.704	0.158	1.84
19.75	560	5.13	1.01	0.197	1.62
20.0	567	5.00	1.14	0.227	1.48
20.25	573	4.98	1.29	0.259	1.35
20.5	580	4.41	1.22	0.276	1.29
20.75	587	3.66	1.12	0.307	1.18
21.0	594	2.72	0.890	0.328	1.12
21.5	607	1.67	0.621	0.373	0.99
22.0	621	1.29	0.541	0.419	0.87
22.5	634	1.20	0.535	0.444	0.81
23.0	648	1.14	0.518	0.456	0.79
23.5	661	0.99	0.467	0.472	0.75
24.0	675	0.826	0.388	0.469	0.76
24.5	688	0.649	0.306	0.471	0.75
25.0	701	0.524	0.242	0.462	0.77

2d va 2e. Grafiklar (Graphics)



$x_{min} = 2.24\,(\mathrm{eV})$	$x_{max} = 2.68\,(\mathrm{eV})$

3c.

$$\alpha h\nu = A(h\nu - E_g)^{\frac{1}{2}} \Rightarrow (\alpha t h\nu)^2 = (At)^2 (h\nu - E_g)$$

$$\Rightarrow y = (At)^2(x - E_g) \Rightarrow m = (At)^2 \Rightarrow t = \frac{\sqrt{m}}{A}$$

$$\Rightarrow \frac{\Delta t}{t} = \frac{\Delta m}{2m}$$

$t = \dfrac{\sqrt{m}}{A}$
$\Delta t = \dfrac{\Delta m}{2A\sqrt{m}}$

Rasm (Grafik): R_{nw} va R_{nw}^{-} ning to'lqin uzunligiga bog'liqligi.

Figure 17: Grafik 2d: Qarshilik to'lqin uzunligi funksiyasi sifatida.

Rasm (Grafik): O'tkazuvchanlik (T_{film}) to'lqin uzunligiga bog'liqligi.

Figure 18: Grafik 2e: Transmission to'lqin uzunligi funksiyasi sifatida.

$$\lambda_{\max}(I_{\text{glass}}) = 564 \pm 5\,\text{nm}$$

$$\lambda_{\max}(I$$

$$\boxed{\text{Grafik: } y \text{ vs } x \text{ (chiziqli soha).}}$$

Figure 19: $y = (\alpha t h\nu)^2$ ning $x = h\nu$ ga bog'liqligi. Chiziqli moslama.

Chiziqli sohadagi x qiymatlari:

$$x_{\min} = 2.24\,\text{eV}, \qquad x_{\max} = 2.68\,\text{eV}$$

3c

$$\begin{aligned} \alpha h\nu &= A(h\nu - E_g)^{1/2} \\ \Rightarrow (\alpha t h\nu)^2 &= (At)^2(h\nu - E_g) \\ \Rightarrow y &= (At)^2(x - E_g) \end{aligned}$$

Bu yerda $y = (\alpha t h\nu)^2$, $x = h\nu$.

Chiziqli moslamaning qiyaligi (slope):

$$m = (At)^2 \quad \Rightarrow \quad t = \frac{\sqrt{m}}{A}$$

Xatolik:

$$\frac{\Delta t}{t} = \frac{\Delta m}{2m} \quad \Rightarrow \quad \Delta t = \frac{\Delta m}{2A\sqrt{m}}$$

$A = 0.071\,(\text{eV})^{1/2}/\text{nm}$ berilgan. m va Δm ni grafikdan topiladi.

3b

Grafik: y (eV^2) vs x (eV) – chiziqli soha. Nuqtalar 530 nm atrofidagi to'lqin uzunliklariga mos keladi.

Figure 20: $y = (\alpha t h\nu)^2$ ning $x = h\nu$ ga bog'liqligi.

Chiziqli sohadagi x qiymatlari:

$$x_{\min} = 2.24 \text{ eV}, \qquad x_{\max} = 2.68 \text{ eV}$$

3c

Asosiy tenglama:

$$\alpha h\nu = A(h\nu - E_g)^{1/2}$$

Ikkala tomonni kvadratga ko'tarib, t ga ko'paytirib:

$$(\alpha t h\nu)^2 = (At)^2(h\nu - E_g)$$

Belgilashlar:

$$y = (\alpha t h\nu)^2, \quad x = h\nu, \quad m = (At)^2$$

U holda:

$$y = m(x - E_g) \quad \Rightarrow \quad \text{to'g'ri chiziq qiyaligi } m = (At)^2$$

$$\Rightarrow t = \frac{\sqrt{m}}{A}$$

Nisbiy xatolik:

$$\frac{\Delta t}{t} = \frac{1}{2}\frac{\Delta m}{m} \quad \Rightarrow \quad \Delta t = \frac{\Delta m}{2A\sqrt{m}}$$

Chiziqli sohada grafik tahlil natijalari:

$$m = 213 \text{ eV}, \quad r^2 = 0.9986, \quad E_g = 2.17 \text{ eV}, \quad A = 0.071 \text{ (eV)}^{1/2}/\text{nm}$$

Shuning uchun:

$$t = \frac{\sqrt{213}}{0.071} \approx 206 \text{ nm}$$

Xatoliklarni hisoblash:

$$\delta x \approx 0.014 \text{ eV}, \quad \delta y \approx 0.9 \text{ (eV)}^2$$

$$\Delta m \approx \sqrt{\frac{(\delta y)^2 + (m\delta x)^2}{\sum_i x_i^2 - N\bar{x}^2}} \approx 10 \text{ eV}$$

$$\Delta t = t \cdot \frac{\Delta m}{2m} \approx 206 \cdot \frac{10}{2 \cdot 213} \approx 5 \text{ nm}$$

$$\Delta E_g \approx \frac{1}{m}\sqrt{\left(\frac{m^2\delta x^2 + \delta y^2}{N}\right)^2 + \left(\frac{\bar{y}}{m}\right)^2 \Delta m^2} \approx 0.02 \text{ eV}$$

3d-jadval (Table 3d). 3-rasm bo'yicha hisoblangan E_g va t qiymatlari

E_g (eV)	ΔE_g (eV)	t (nm)	Δt (nm)
2.17	0.02	206	5